# Main (géo-)visualisation approaches

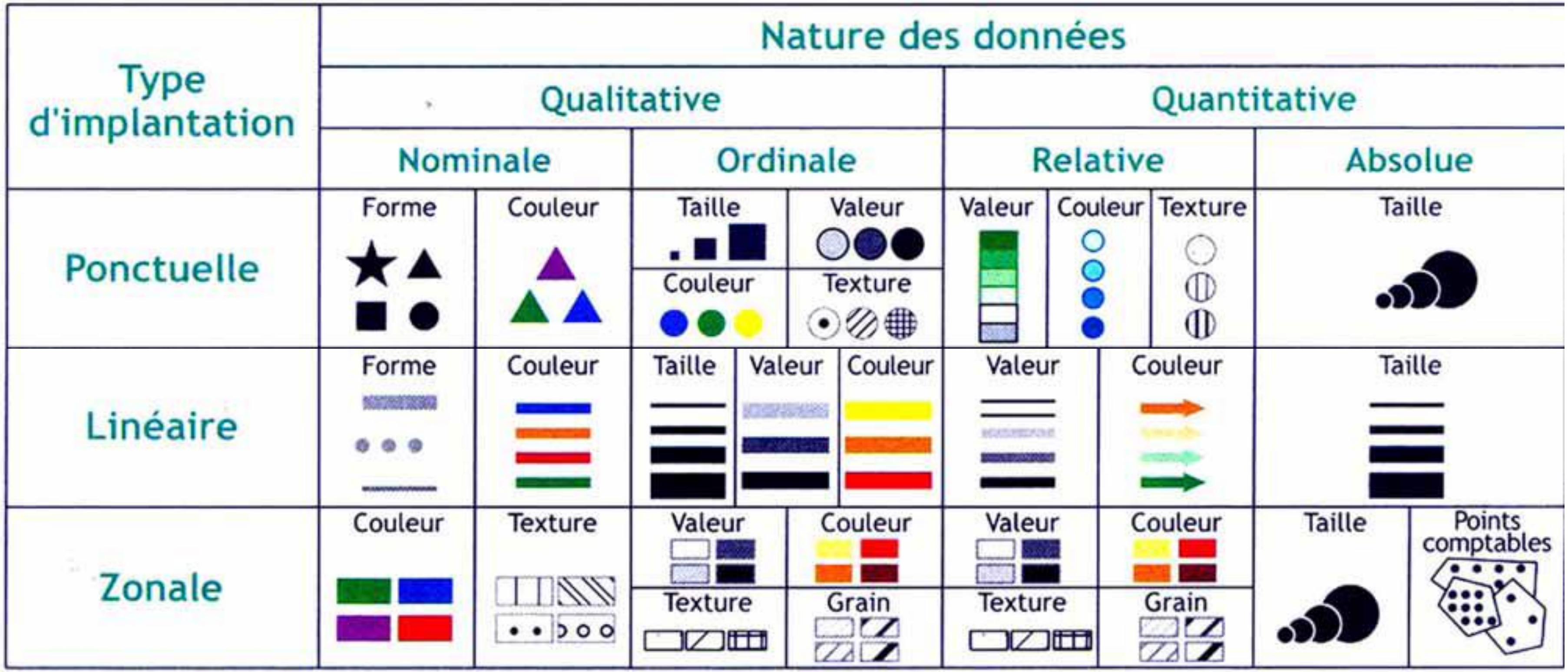
**Jacques Gautier**

IGN-ENSG, LaSTIG, Equipe GeoVIS, DCAIG

# Rules of thumbs in symbology
# Focus on cartography

# Using adequate visual encodings to represent data

Zanin, Trémolo. Savoir faire une carte. 2003.



# Munzner visualization taxonomy

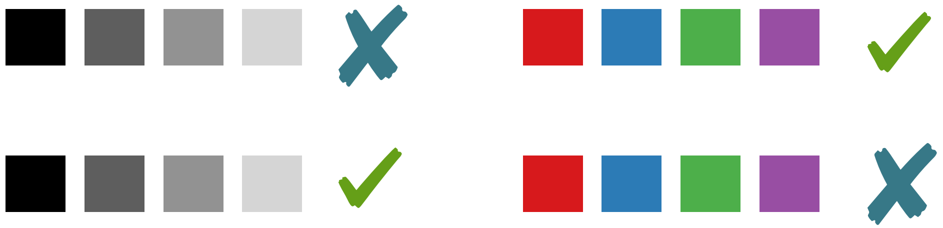
## Expressiveness *(Munzner, 2014)*

“visual encoding should express all of and only the information in the dataset.”

- → Ordered data should be represented in a way that is perceived as ordered
- → Unordered data should not be represented in a way that is perceived as ordered

Countries

Population



# Munzner visualization taxonomy

## Effectiveness *(Munzner, 2014)*

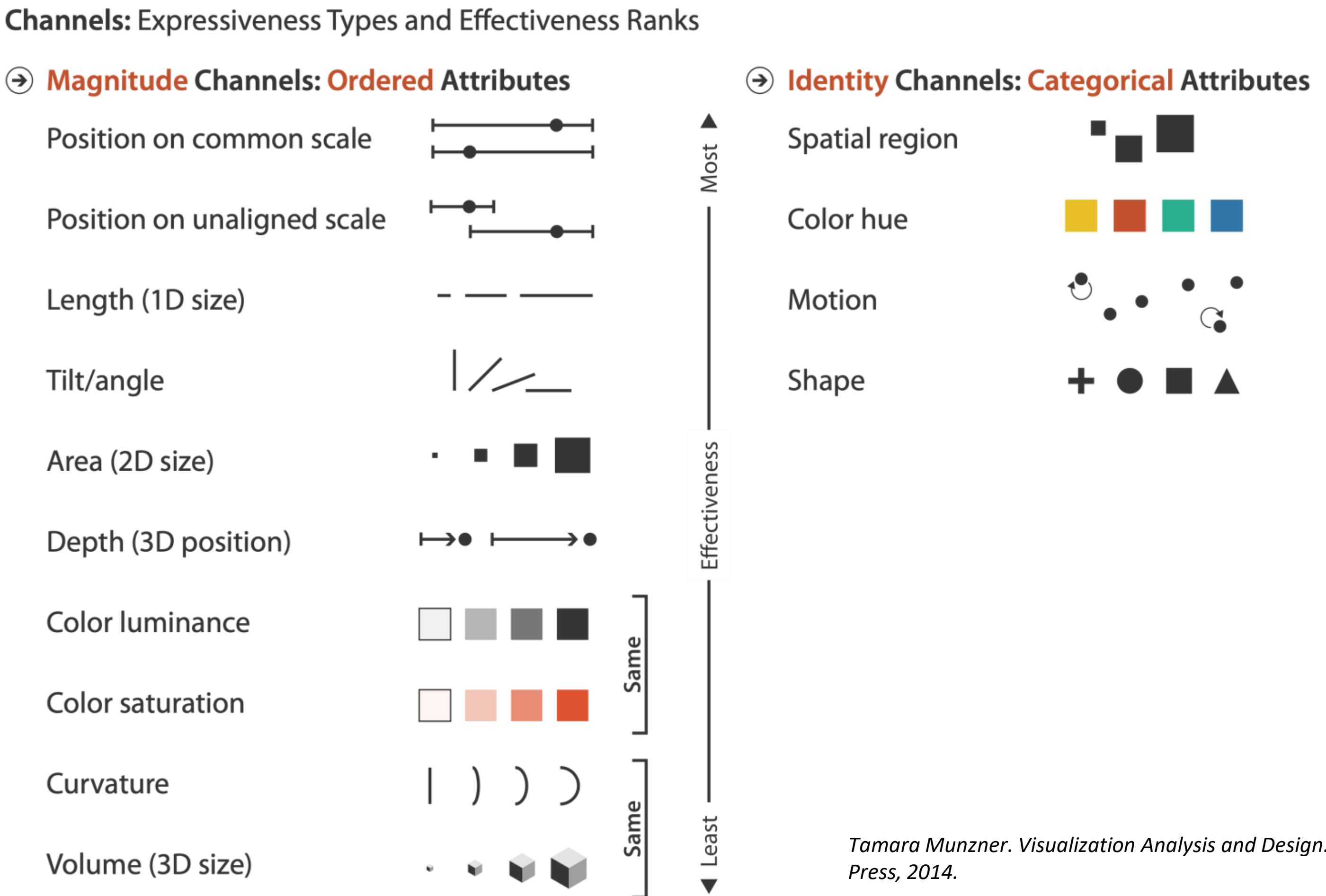
“Channels differ in accuracy of perception.”

→ Most important attributes should me encoded with the more effective channels

Less important

Most important

# Visual Channels




*Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014.*

# How is effectiveness evaluated?

**Accuracy** : With what precision can the value represented by a visual channel be evaluated?

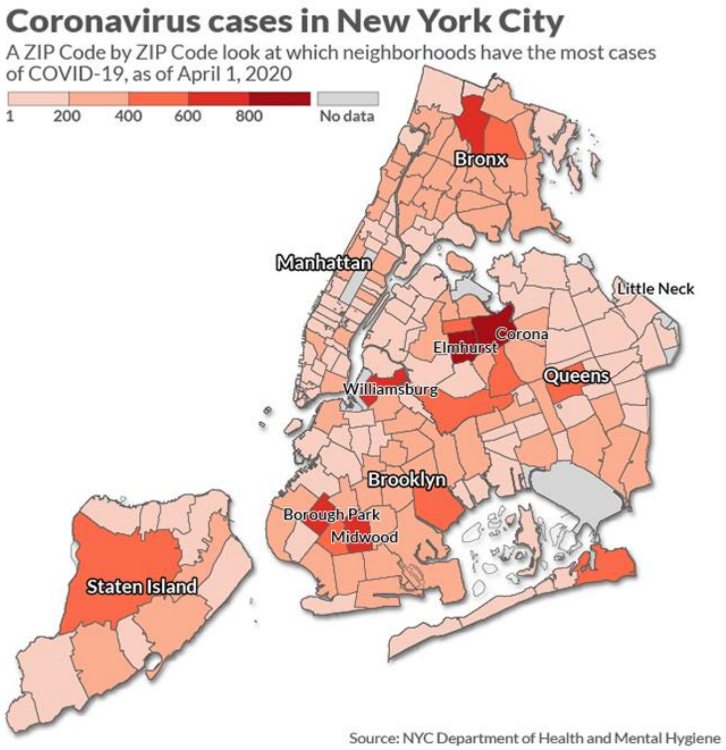
**Discriminability** : How many different modalities of the same visual channel are perceptible?

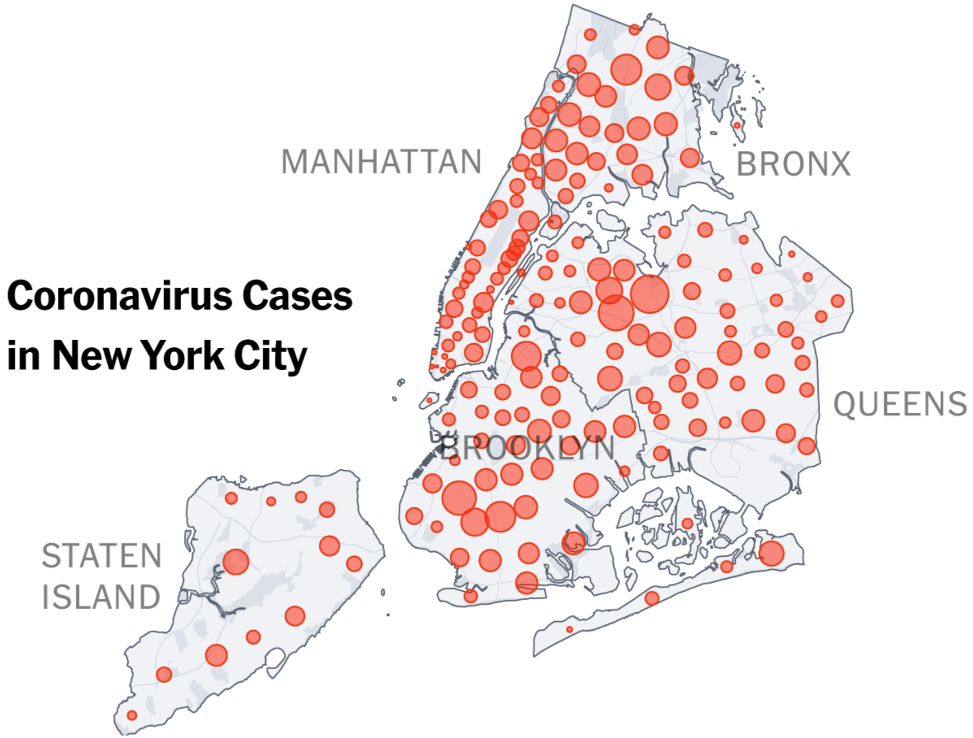
**Separability** : In a representation using two visual channels, is it possible to separately distinguish the modalities of these two visual channels?

*Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014.*

# One of the most important rule in cartography

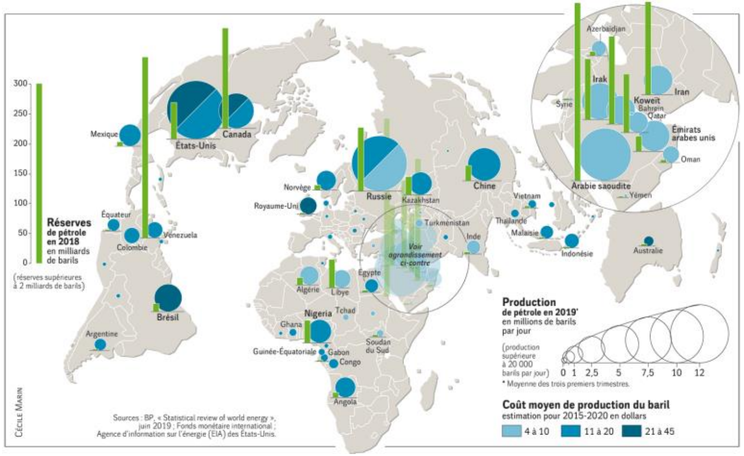
**Never** represent a absolute quantity with color



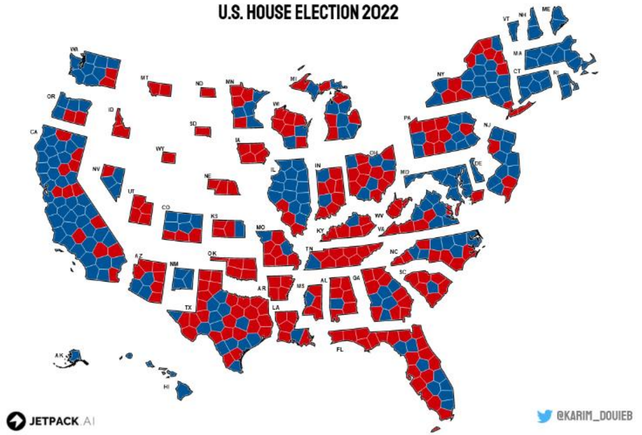



# Representing quantities

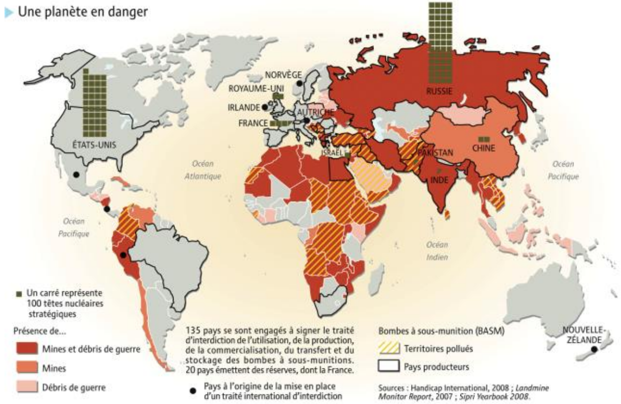
https://observablehq.com/@karimdouieb/us-house-election-2022

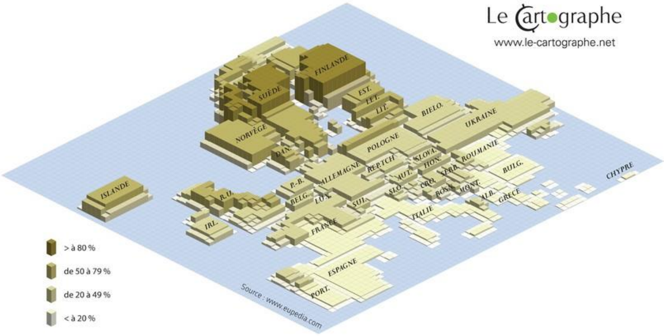



Production and consumation of oil
Monde Diplomatique
https://www.monde-diplomatique.fr/cartes/petrole-prod-conso



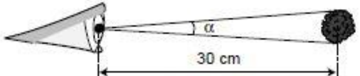

A planet in danger
Une planète en danger
Atlas du Monde Diplomatique, Un Monde à l'Envers, 2009,
https://www.monde-diplomatique.fr/cartes/l_atlas_un_monde_a_l_envers/a60829



# Don’t use too many channel modalities

DIMENSIONS GRAPHIQUES MINIMALES

| CONTRAINTES VISUELLES | REPRÉSENTATION GRAPHIQUE |
| --- | --- |
| Acuité visuelle de discrémination |   A = 1 minute sexagésimale = 0,09mm = 1/10 mm |
| Acuité visuelle d'alignement | 0,02mm |
| Seuil de perception | Ponctuel ● 0,2 mm ■ 0,4 mm ○ 0,3 mm □ 0,5 mm △ 1 mm 0,5 mm 0,6 mm<br>Linéaire 0,1 mm |
| Seuil de séparation | Linéaire 2/10 3/10<br>Ponctuel 2/10 |
| Seuil de différenciation | Ponctuel ● ● entre 2 paliers le rapport des surfaces doit être au moins de 2<br>Linéaire Traits rapprochés ecart d'épaisseur 0,1mm<br>Traits éloignés 0,3 mm minimum |

Beware of symbols discrimination capacities of the user

Source: General cartography course, Gérald Weger, ENSG, Mars 1999

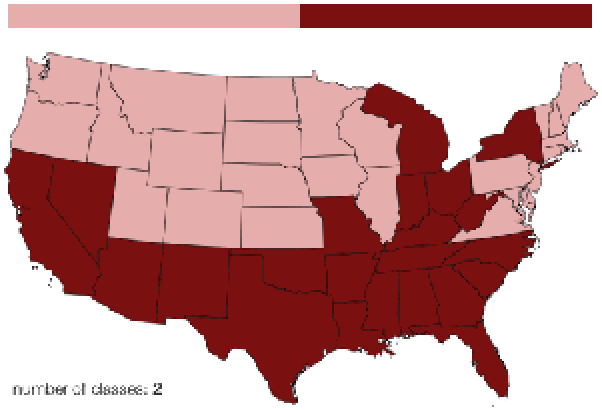
# Color discrimination

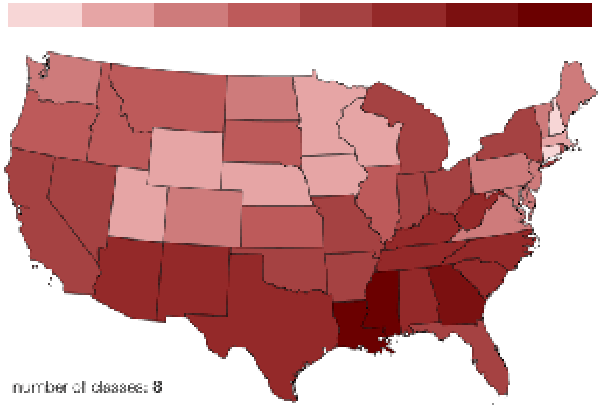
Rule of thumb: 6-12 bins, including background and highlights

Color discrimination:

- great if color contiguous
- bad for absolute comparisons

For non contiguous regions of color, we have fewer bins than we want




*Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014.*

*Gregor Aisch, vis4.net/blog/posts/choropleth-maps/*

# Color discrimination

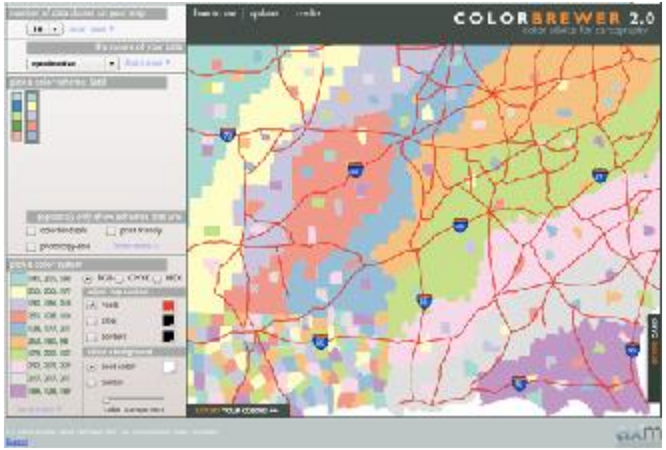
## color channel interactions

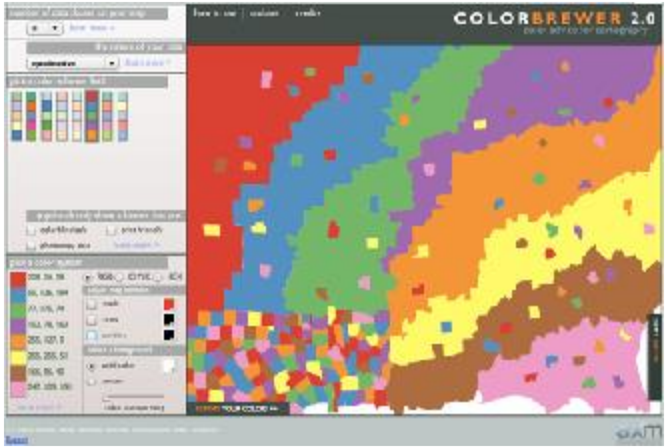
Size heavily affects salience

- small regions need high saturation
- large regions need low saturation
- small separated regions: 2 bins safest (use only one of these channels), 3-4 bins max

## saturation & luminance

- not separable from each other
- not separable from transparency



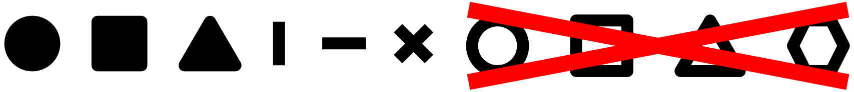



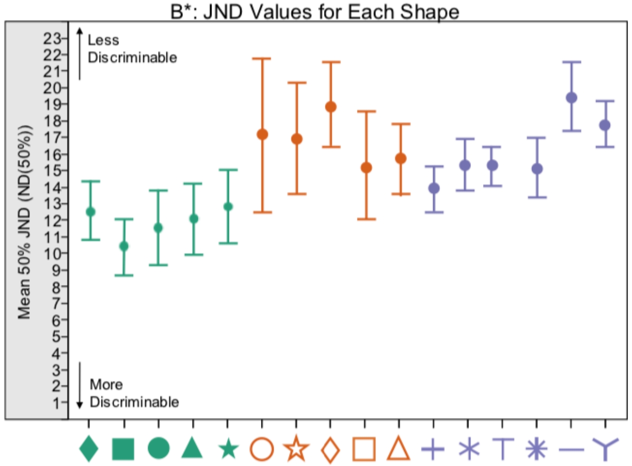

*http://colorbrewer2.org/*

*Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014.*

# Using shapes

Avoid using too many different shapes
Avoid empty geometric shapes: symbol colors are les discriminable






Smart, Stephen et Danielle Albers Szafir. "Mesurer la séparabilité de la forme, de la taille et de la couleur dans les nuages de points." Actes de la conférence CHI 2019 sur les facteurs humains dans les systèmes informatiques. 2019.

# Other rules of thumbs

Never forget

- Map orientation
- Map scale
- Data sources
- **Legend**

# Map legend

Classify information by order of priority

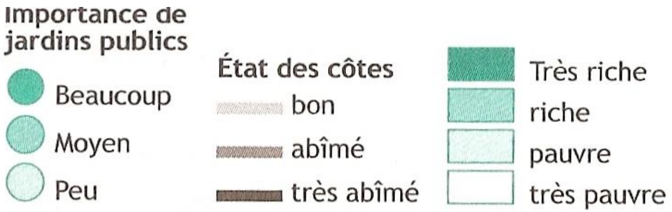
Put the legend elements at a same place in the map

Round data value at a pertinent precision

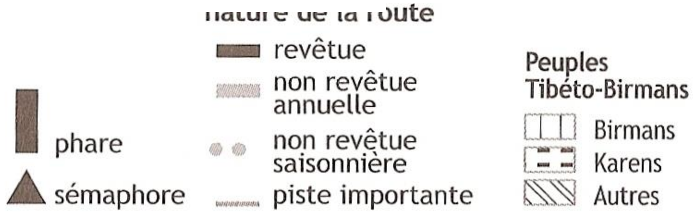
Show min and max values of statistical series

# Map legend

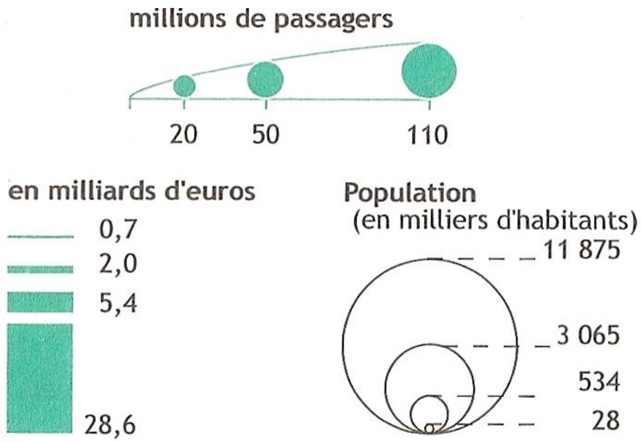
Ordinate qualitative



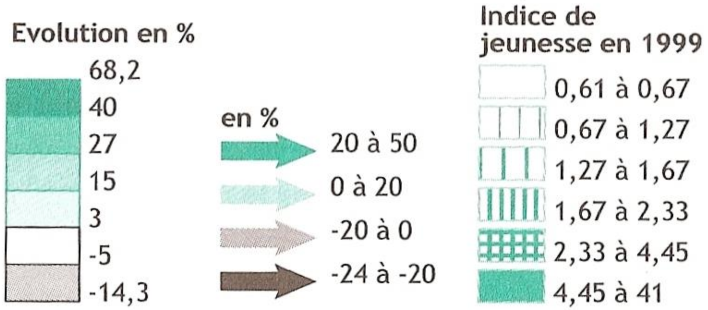

Nominal qualitative




Absolute quantitative




Relative quantitative



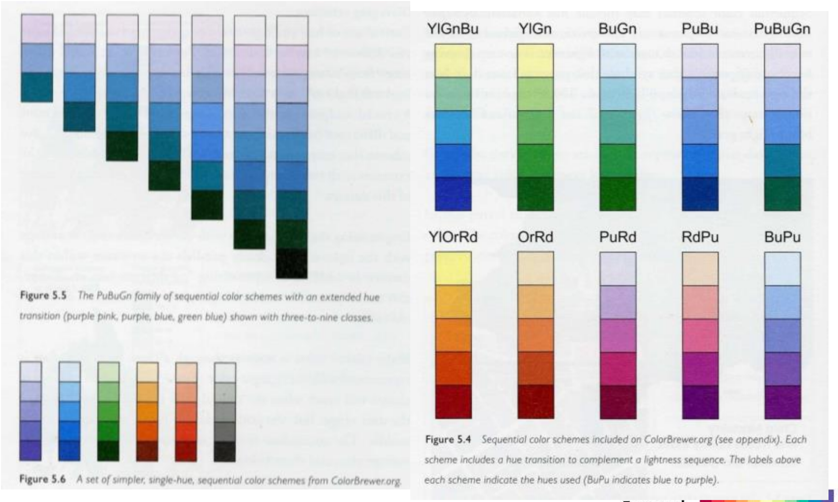

Zanin, Trémolo. Savoir faire une carte. 2003.

# Which color palette?

## **Sequential** palettes

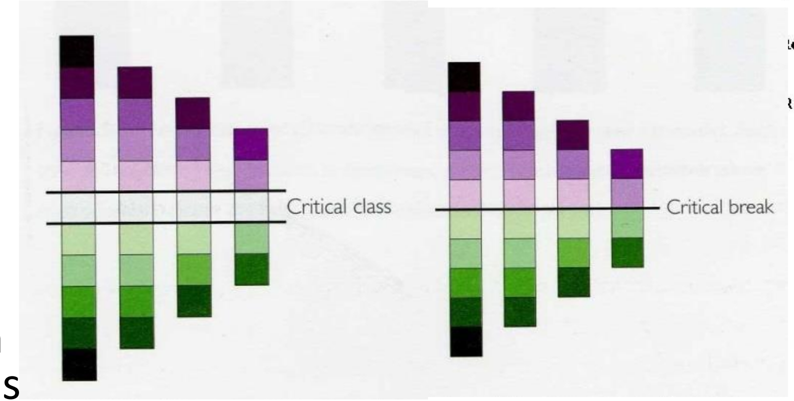
Evoluation of **luminance** or **saturation**

Can use **several hues** (variation of **luminance**)
=>increase the number of discriminable modalities




## **Diverging** palettes

**Double variation** of **luminance/saturation**
Neutral color in the middle (white, pale yellow, grey)
Ordered relationship with a neutral median value between two opposite extreme values

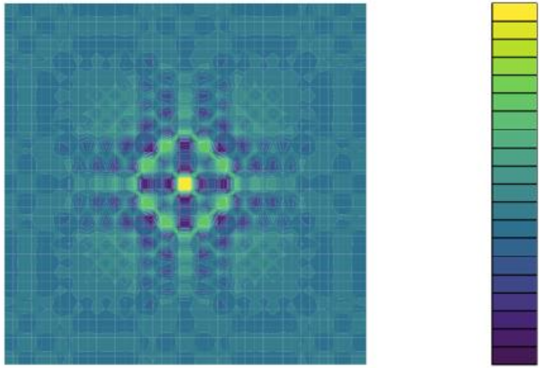


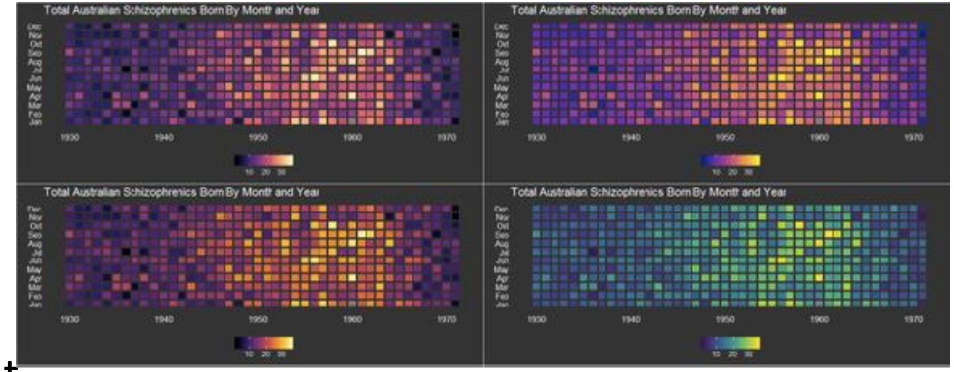


*after [Color Use Guidelines for Mapping and Visualization. Brewer, 1994. http://www.personal.psu.edu/faculty/c/a/cab38/ColorSch/Schemes.html]*

# Viridis and Magma

- Linear gradient of luminance
- Uniform perception along the palette
- Colorblind-safe

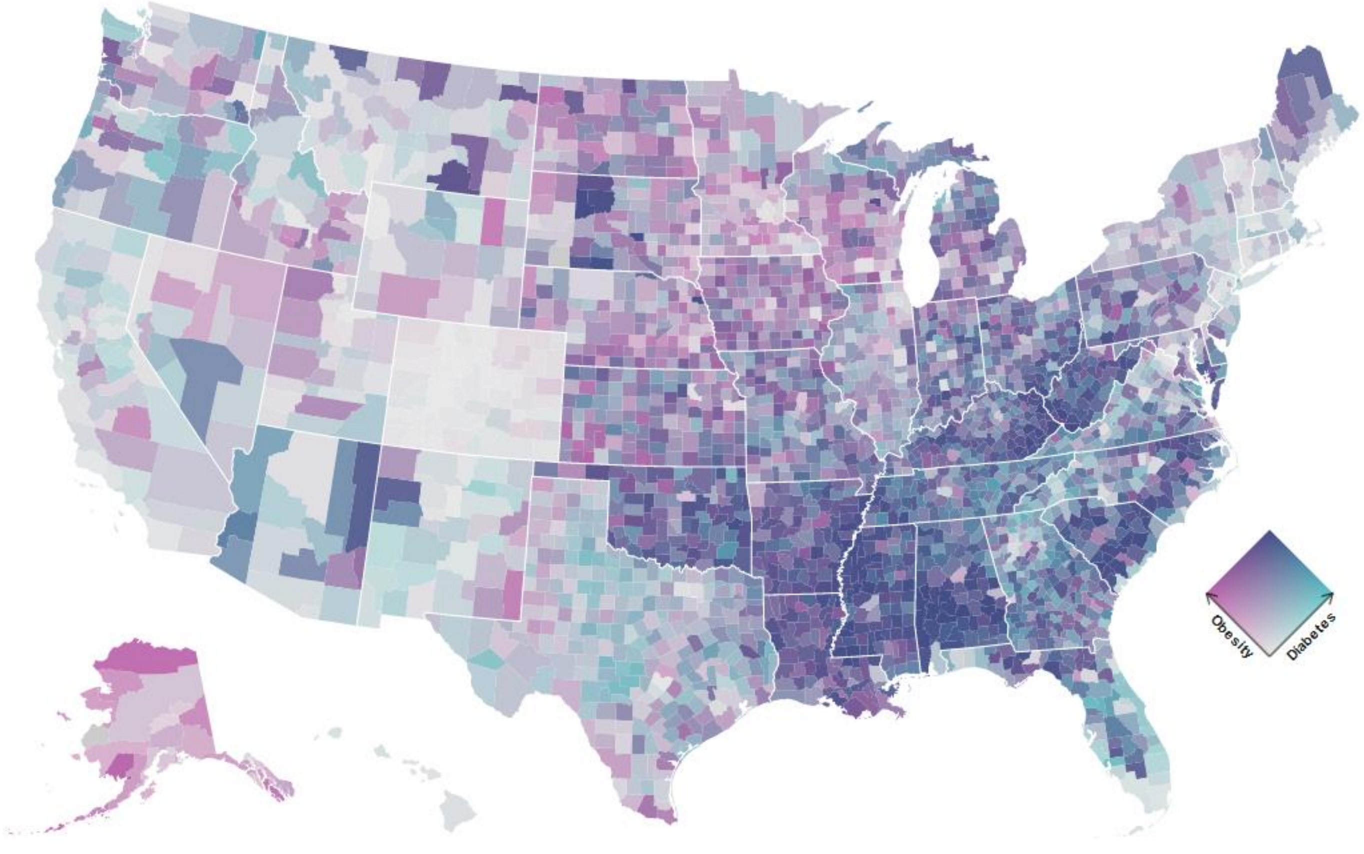





https://cran.r-project.org/web/packages/viridis/vignettes/intro-to-viridis.html

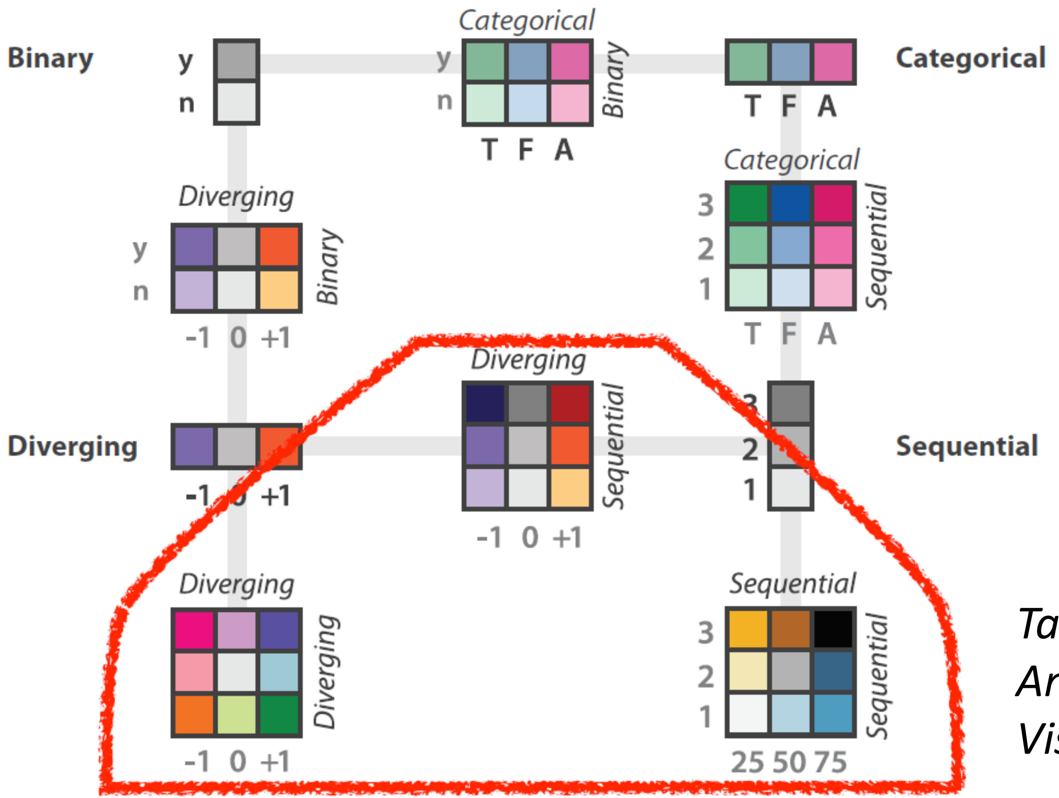
*Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014.*

# Bivariate color palettes




https://observablehq.com/@stephanietuerk/bivariate-choropleth-with-continuous-color-scales

# Bivariate color palettes




Bivariate color scales can be very difficult to read in case of numerous modalities on two sequential axes

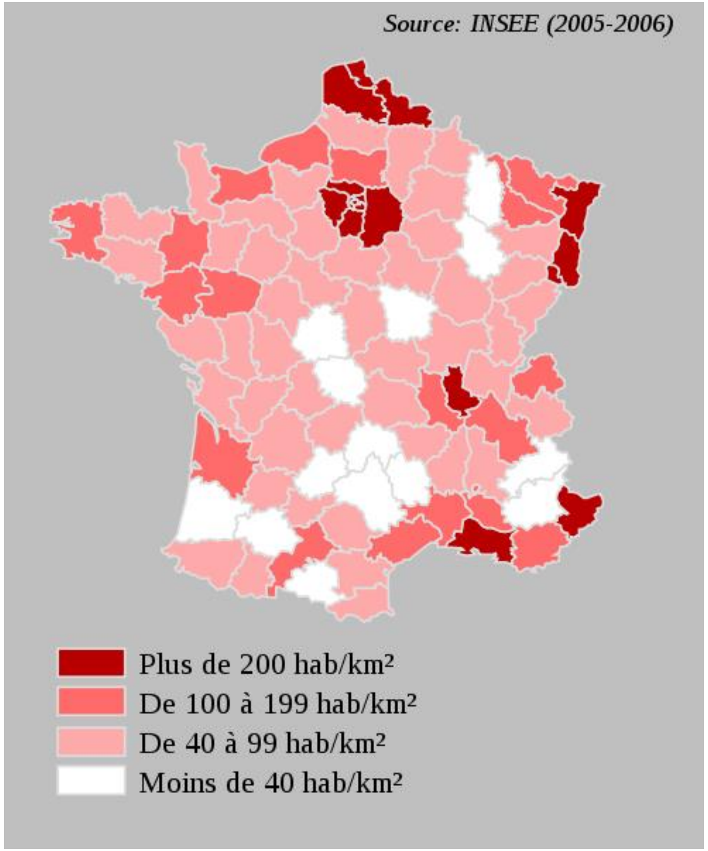
*Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014.*

# Discrete color scales

Choropleth map

Representation of quantitative values (plethos) …
… related to geographic areas (khorê)…
… using a graduated value scale which represents ordered classed values

**Importance of the discretization /classification method**




Map of population density, Wikipedia

https://observablehq.com/@d3/quantile-quantize-and-threshold-scales

# Discrete color scales

Data

**Quantile**: Each class has the same number of individuals

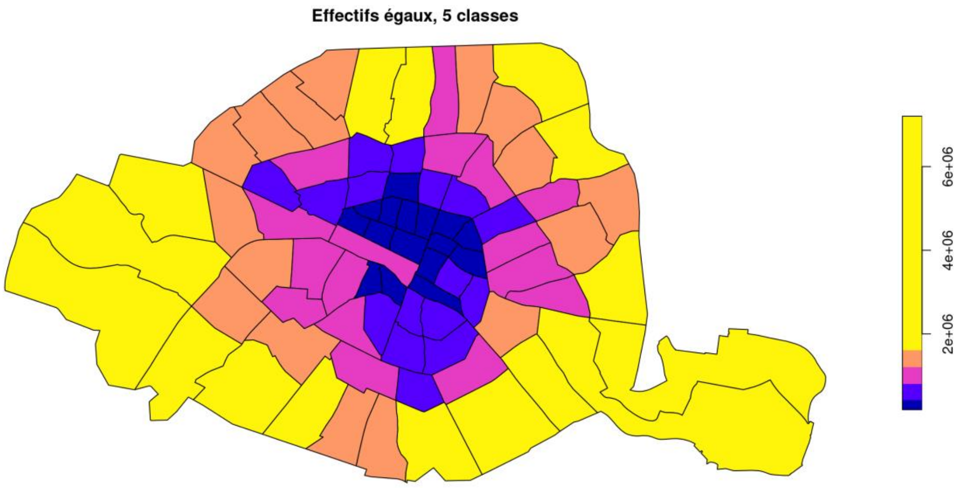
**Quantize**: Each class cover equal value intervals

**Jenks natural breaks** classification method:
Seeks to reduce the variance within classes and maximize the variance between classes

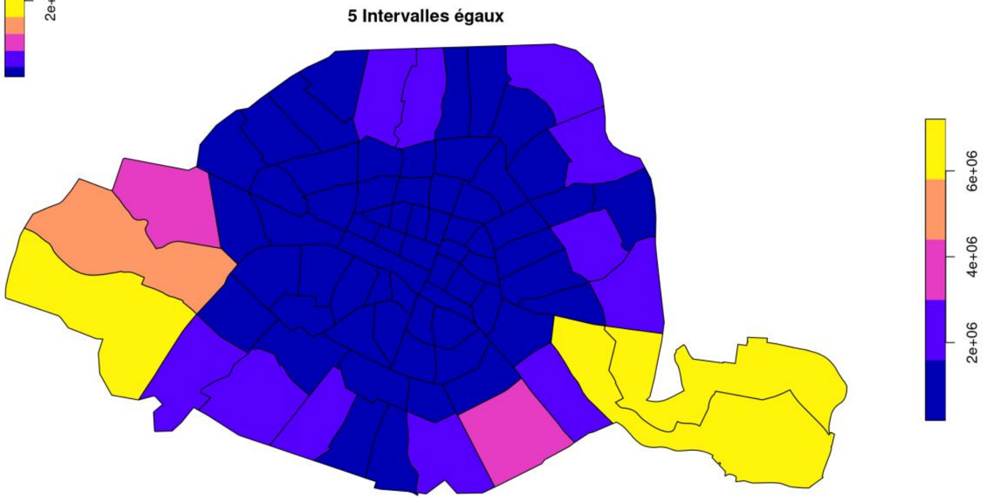
**Manual thresholds**: classes limits are determined manually by the map conceptor according to data distribution

https://www.umr-lastig.fr/paul-chapron/resources/cours_site/carto-discretis.html

# Discrete color scales




Quantile




Quantize

# Taking the user into account

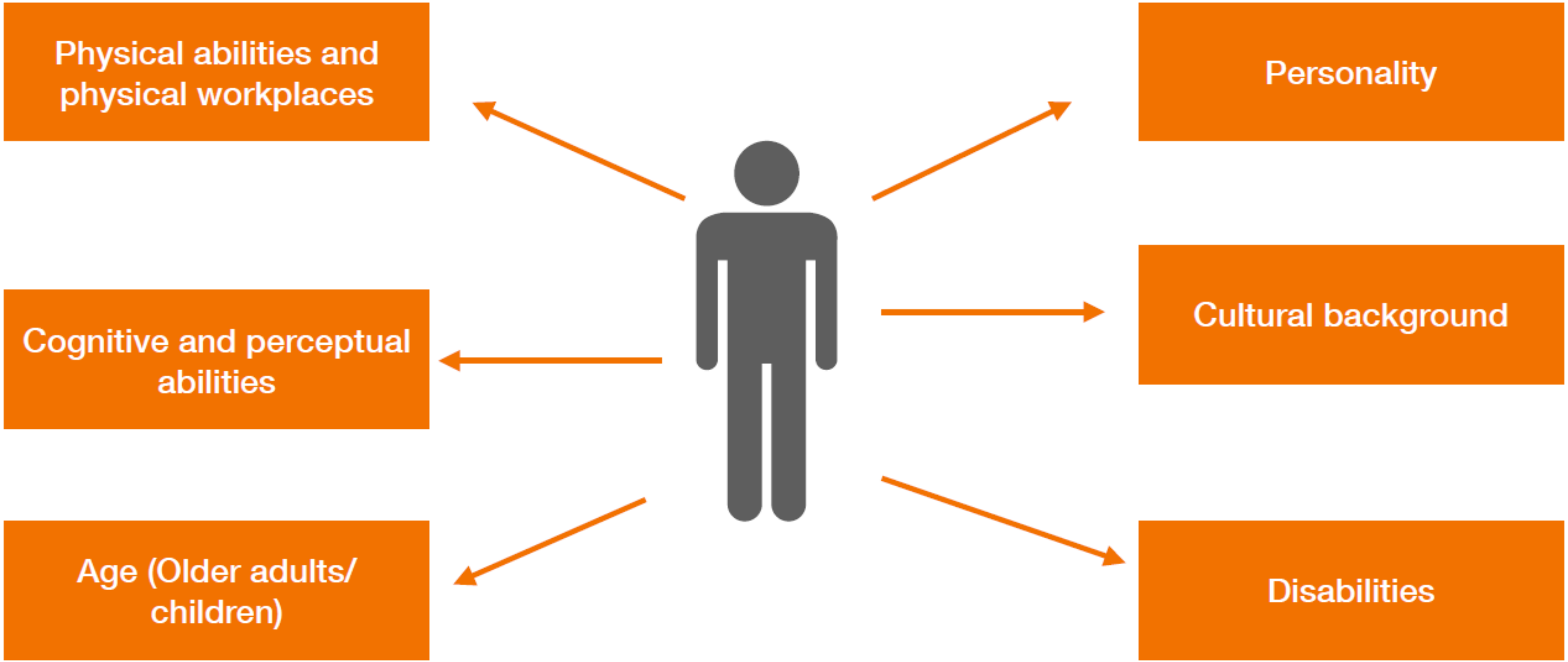
# Which approach for conception

Always keep in mind

- **For which purpose** the visualization is conceived
- **Who** will use it
- **In which context** it will be used

# Which approach for conception

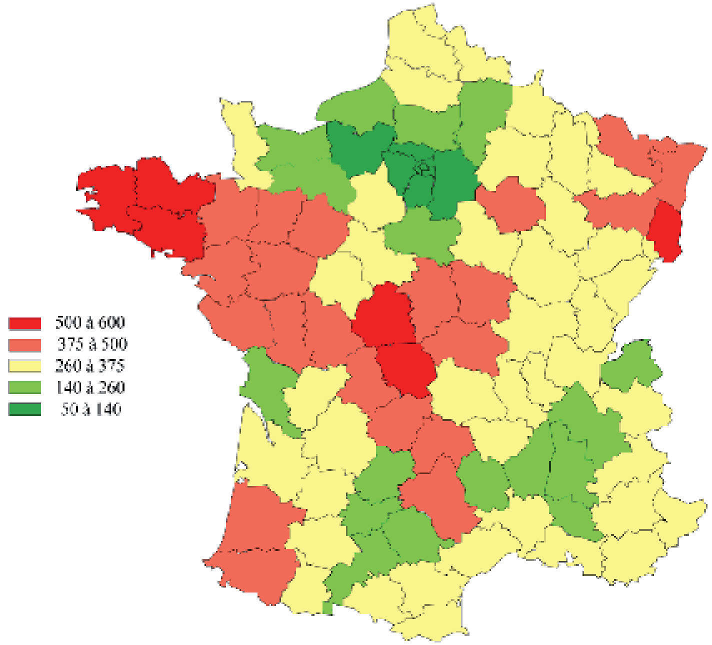
- Which users?



# Which users ?

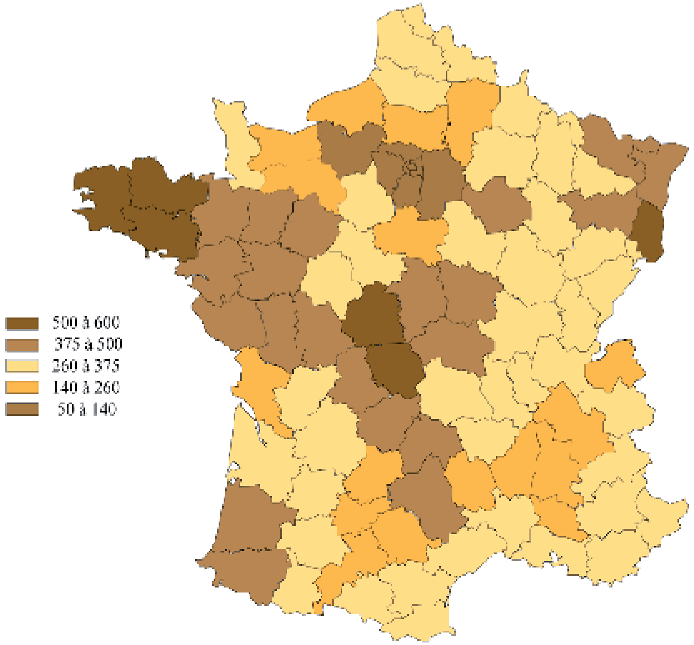
## Dyschromatopsia: total or partial color perception anomaly

Taux de pénétration des quotidiens de province




Nombre de quotidiens diffusés pour 1000 foyers en 1996

Taux de pénétration des quotidiens de province




Nombre de quotidiens diffusés pour 1000 foyers en 1996

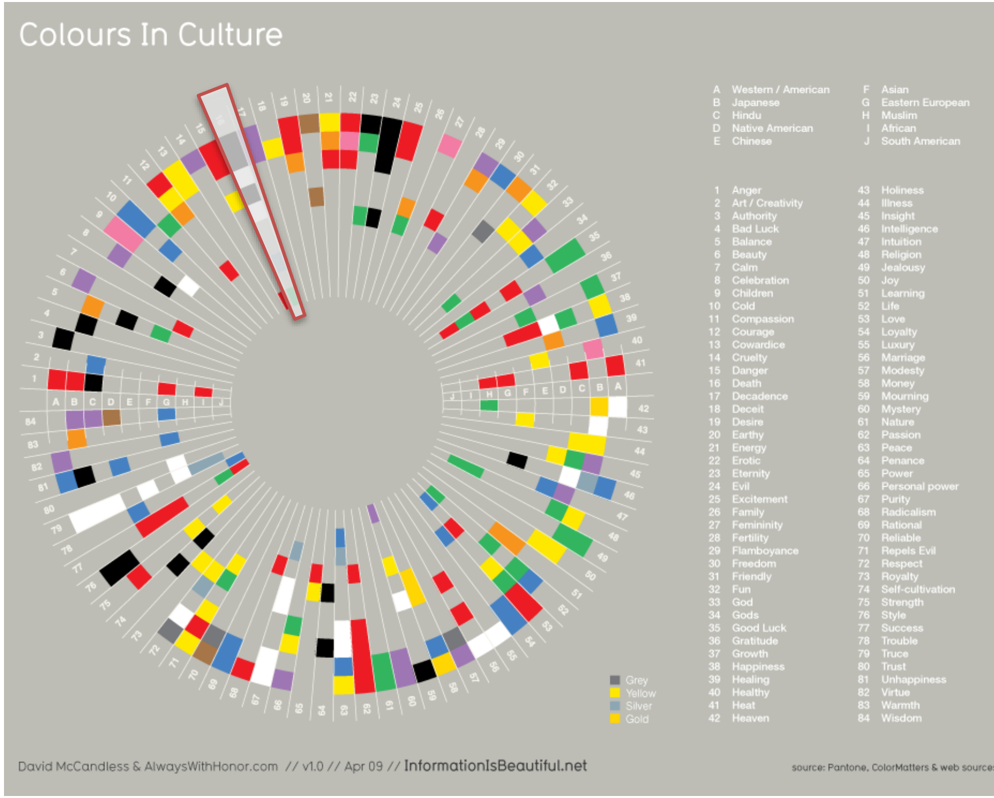
Francis Dhee

# Which users ?

Some colors are used by convention, following cultural relationship with specific concepts

- Hydrographic data: blue
- Vegetation data : green

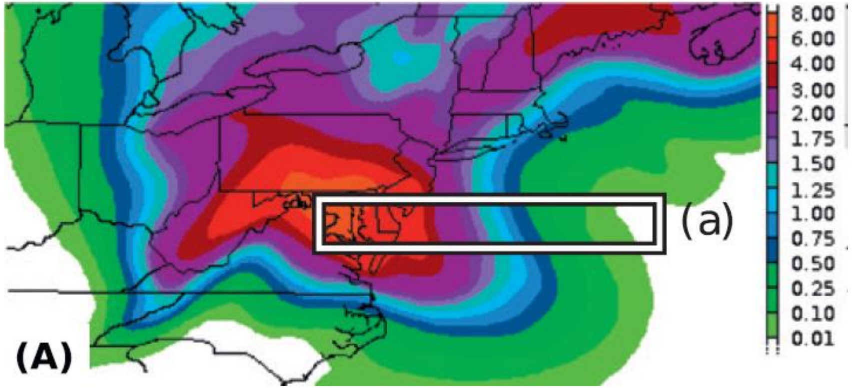
According to cultural background, colors are not related to the same concepts and ideas




www.informationisbeautiful.net/visualizations/colours-in-cultures/

# Which users ?

A rainfall amount forecast during the landfall of Hurricane Sandy on 29 Oct 2012, over the U.S. East Coast.
Somewhere over the rainbow, Stauffer, 2015




« Rainbow » color scales **Widely used in scientific domains (ex: meteorology)**

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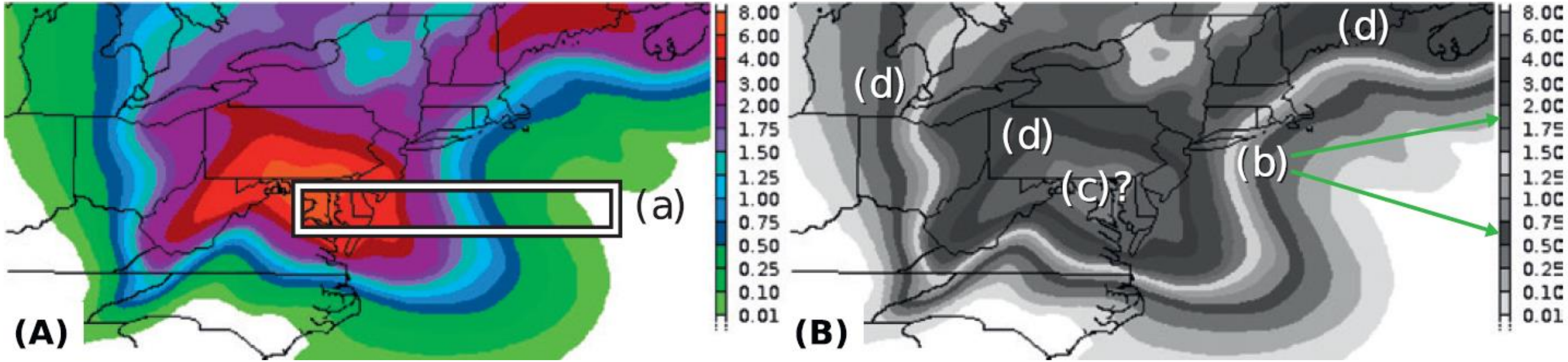


« Rainbow » color scales **Widely used in scientific domains (ex: meteorology)**
**Very criticized in cartography**

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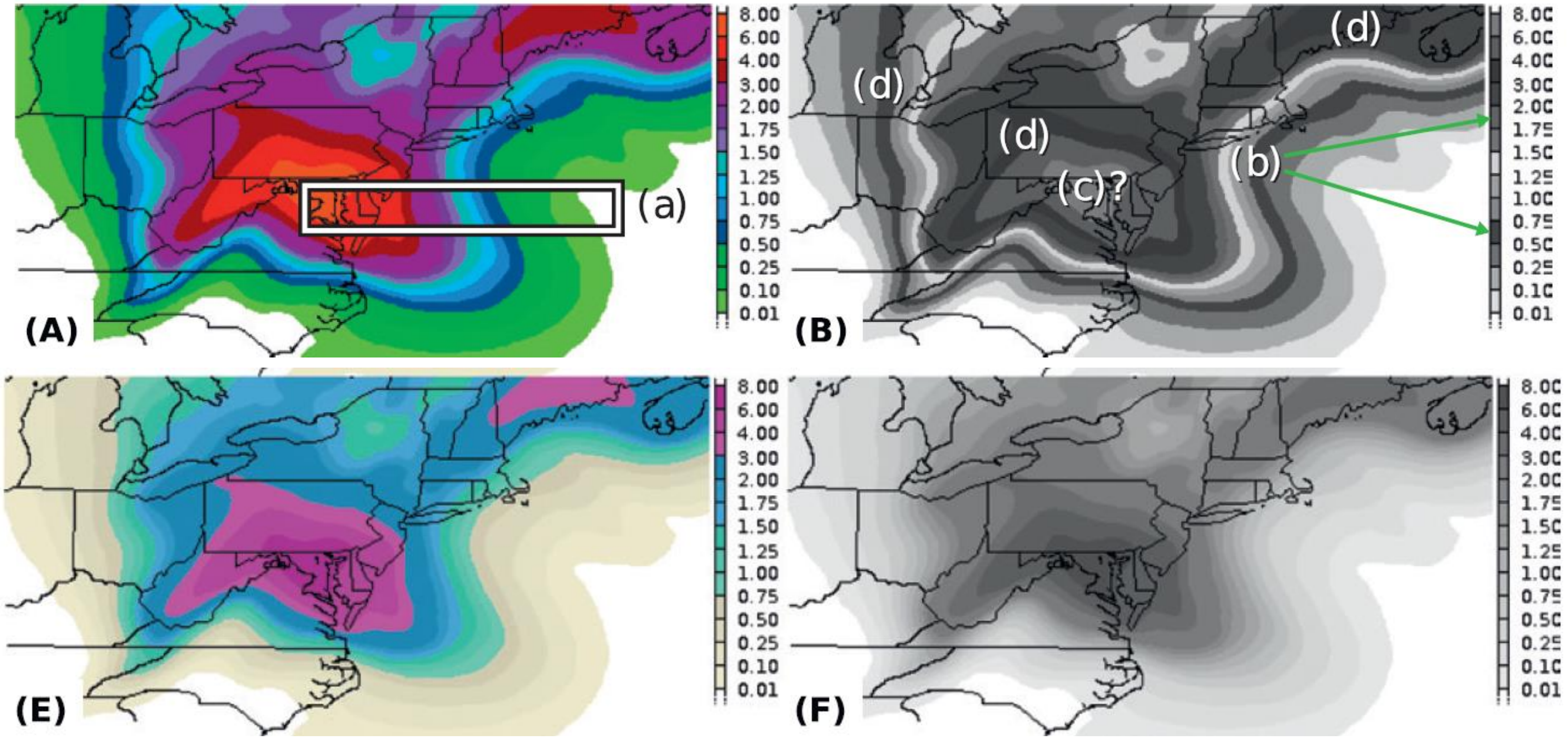
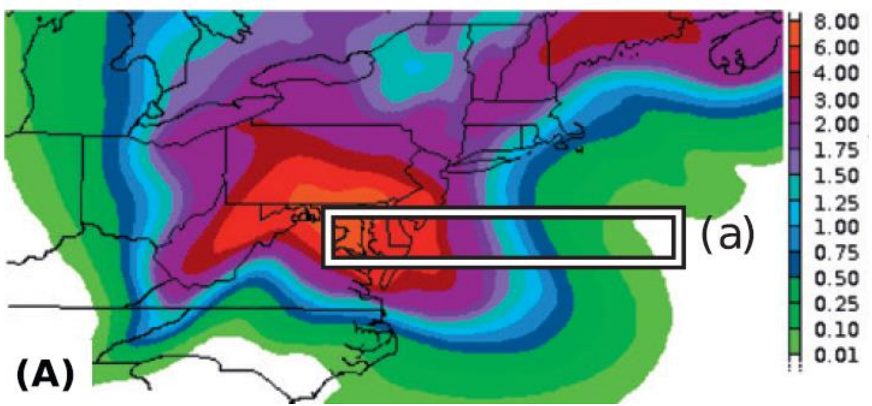


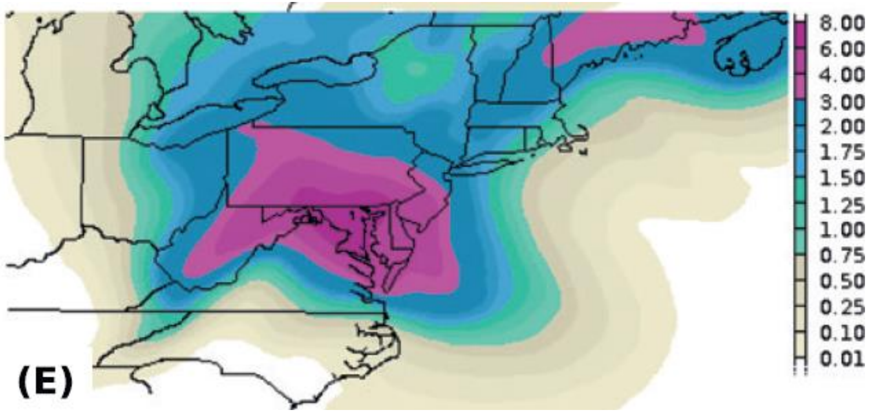

« Rainbow » color scales **Widely used in scientific domains (ex: meteorology)**

**Very criticized in cartography** => *new scales have been explored*

# Which users ?




A change in hue is more effective for identify spatial distribution of different values.
[Dasgupta et al., 2015]




A continuous change in luminance is more effective for interpreting spatial gradients
[Dasgupta et al., 2015]

Dasgupta, A., Poco, J., Rogowitz, B., Han, K., Bertini, E., Silva, C. T., 2018. The effect of color scales on climate scientists' objective and subjective performance in spatial data analysis tasks. IEEE transactions on visualization and computer graphics, PP, 1-1.
Stauffer, R., Mayr, G. J., Dabernig, M., & Zeileis, A. (2015). Somewhere over the rainbow: How to make effective use of colors in meteorological visualizations. Bulletin of the American Meteorological Society, 96(2), 203-216.
Ware, C., Stone, M., & Szafir, D. A. (2023). Rainbow colormaps are not all bad. IEEE Computer Graphics and Applications, 43(3), 88-93.

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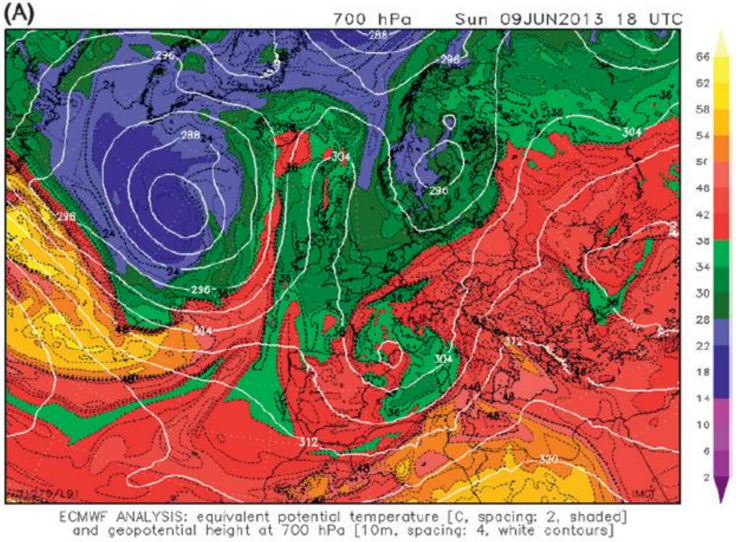


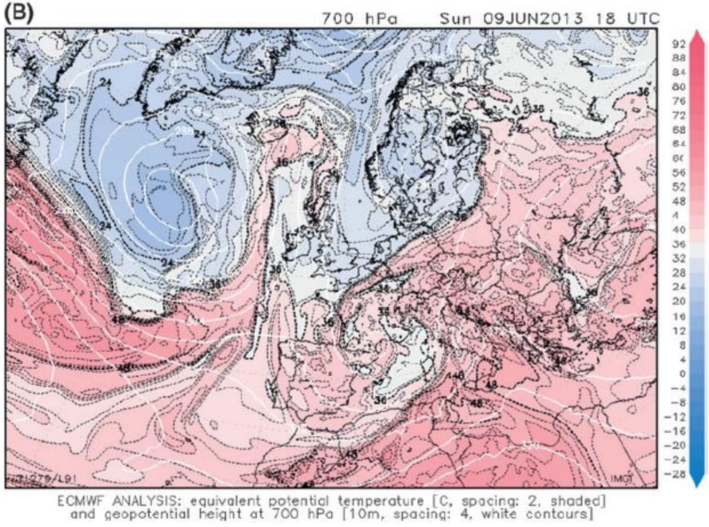


Les échelles divergentes, combinant variation de luminance et variation entre deux teintes opposées, sont plus efficaces pour l'analyse des magnitudes et de la distribution spatiale [Stauffer et al., 2015]

Dasgupta, A., Poco, J., Rogowitz, B., Han, K., Bertini, E., Silva, C. T., 2018. The effect of color scales on climate scientists' objective and subjective performance in spatial data analysis tasks. IEEE transactions on visualization and computer graphics, PP, 1-1.
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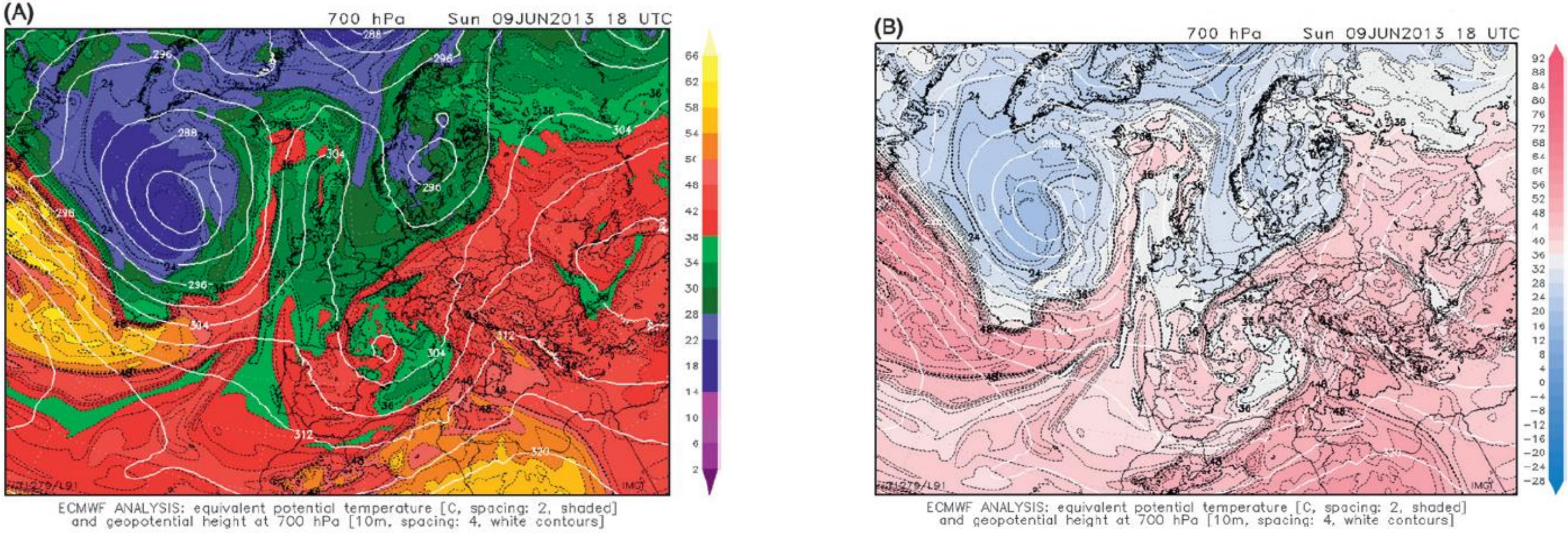


Malgré de moins bons résultats, beaucoup d’utilisateurs experts expriment une meilleure confiance dans les échelles *arc-en-ciel* [Dasgupta et al., 2015]

Le fort contraste apporté par les différentes teintes de couleur permet d’identifier plus facilement des valeurs extrêmes [Ware et al., 2023]

Dasgupta, A., Poco, J., Rogowitz, B., Han, K., Bertini, E., Silva, C. T., 2018. The effect of color scales on climate scientists’ objective and subjective performance in spatial data analysis tasks. IEEE transactions on visualization and computer graphics, PP, 1-1.

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# Which approach for conception

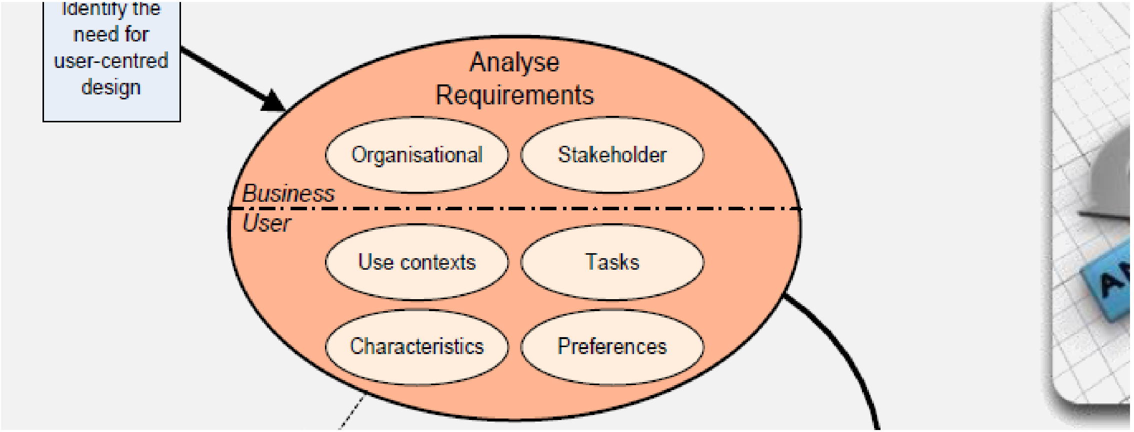
Always keep in mind

- **For which purpose** the visualization is conceived
- **Who** will use it
- **In which context** it will be used

- Never take for granted that the visualization you implement can be correctly used/interpret by everyone

- Adopt a user-centered design approach when conceiving a visualization

# User-Centered Design






*Material from Kristien Ooms - www.univie.ac.at/icacomuse/ http://cartogis.ugent.be/kooms/UUI/*

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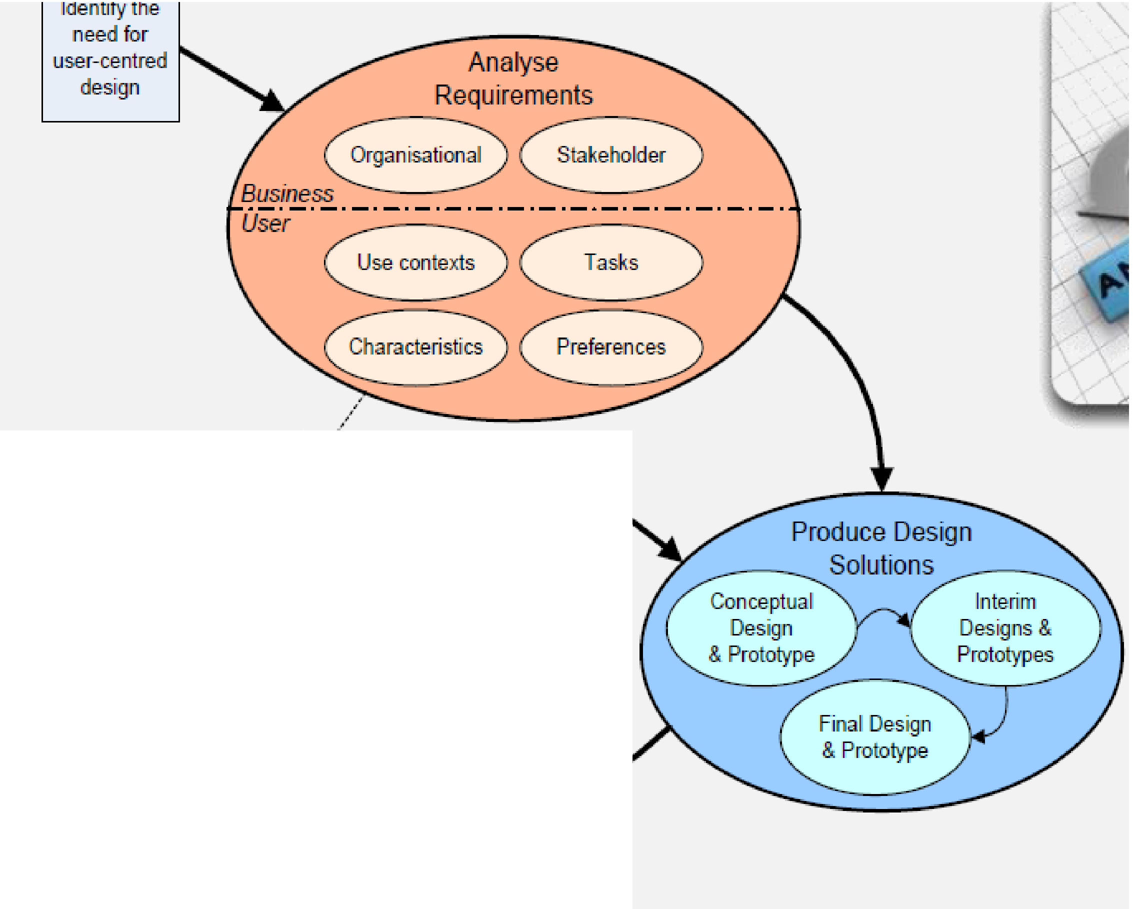


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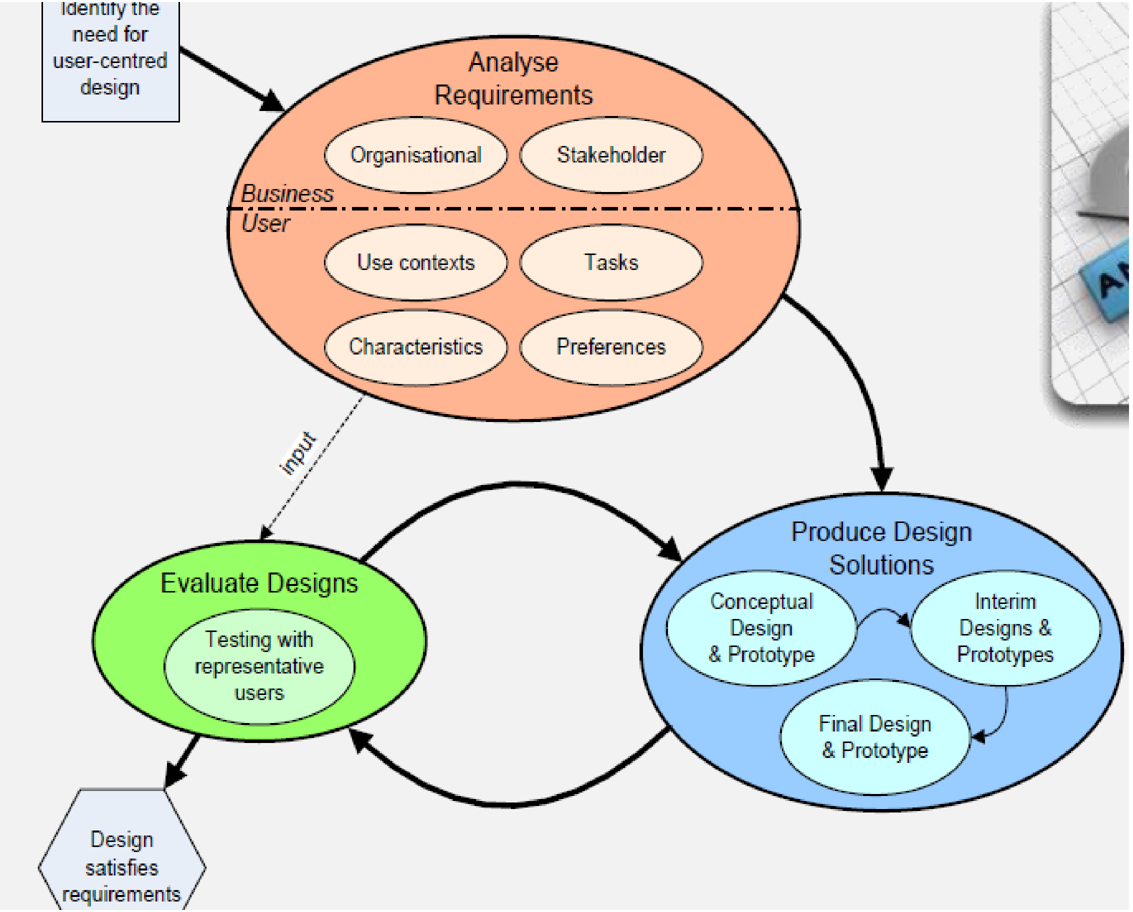


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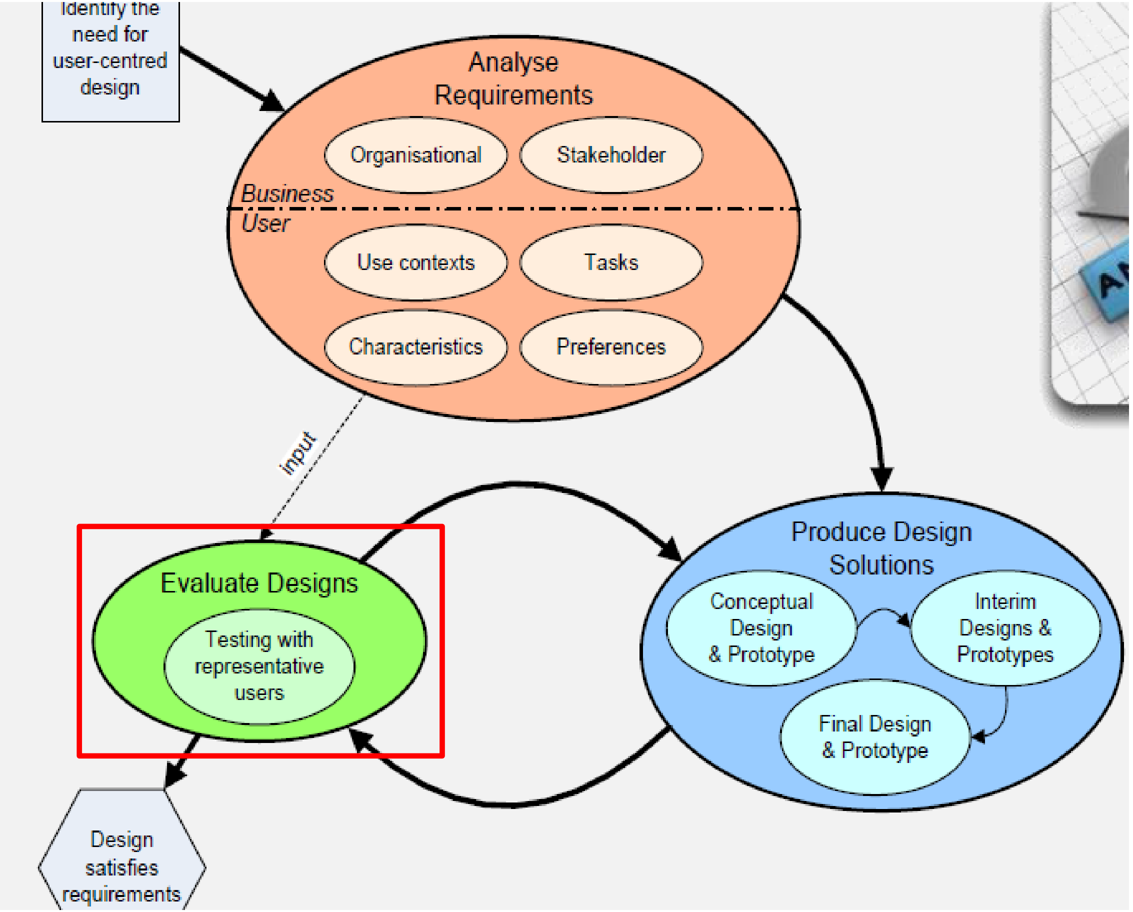


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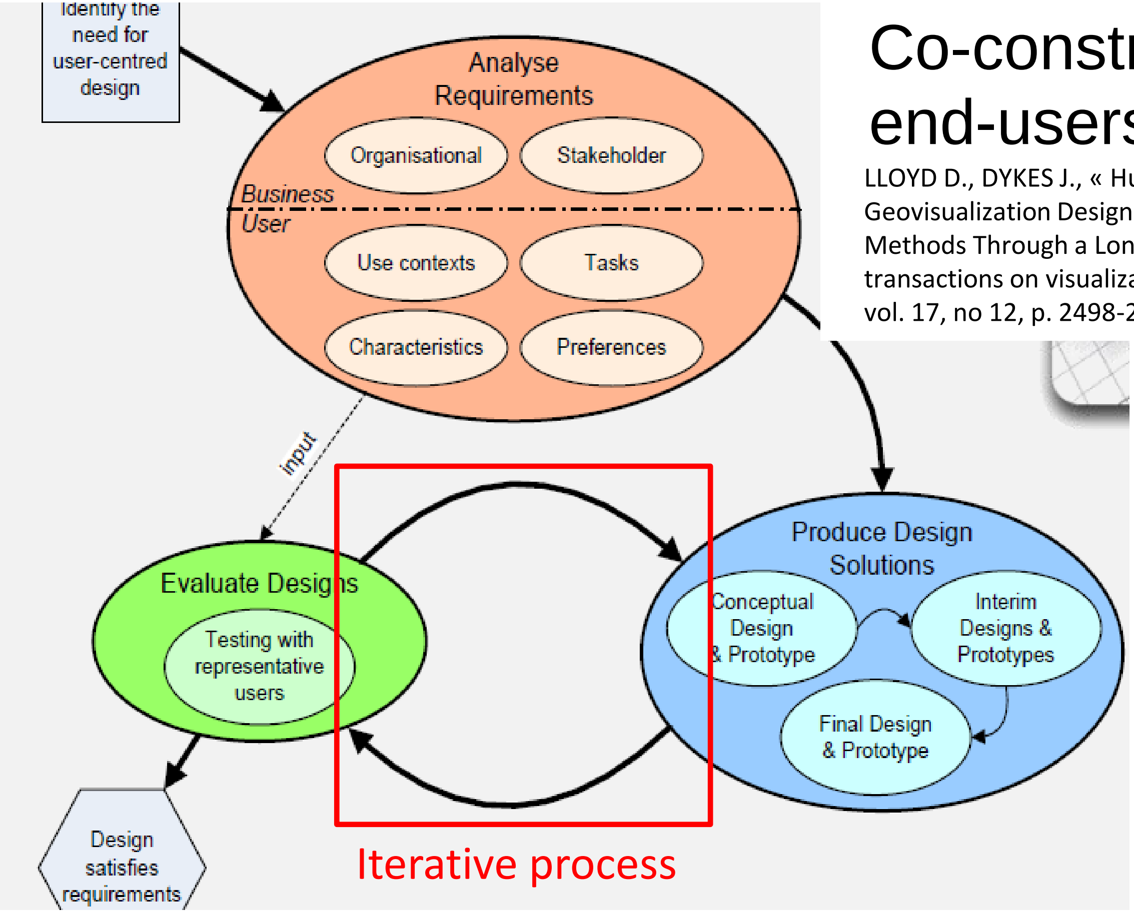
# User-Centered Design




## Co-construction with end-users

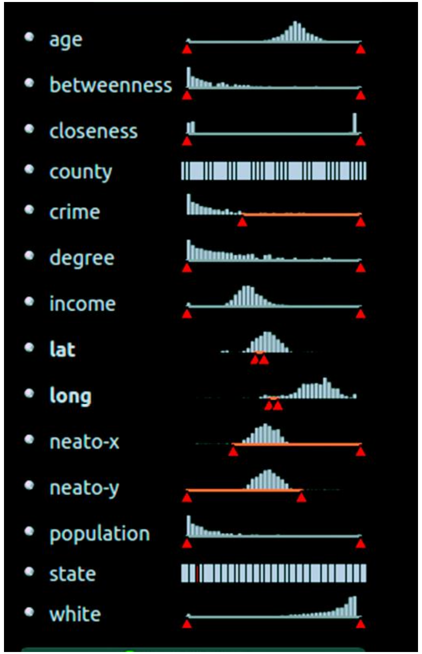
LLOYD D., DYKES J., « Human-Centered Approaches in Geovisualization Design: Investigating Multiple Methods Through a Long-Term Case Study », IEEE transactions on visualization and computer graphics, vol. 17, no 12, p. 2498-2507, 2011.

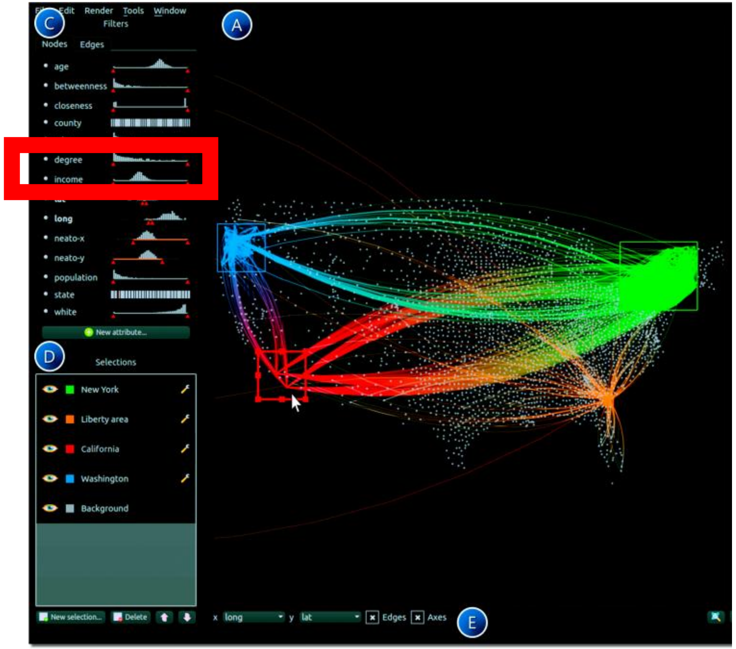
*Material from Kristien Ooms - www.univie.ac.at/icacomuse/   http://cartogis.ugent.be/kooms/UUI/*

# Other approaches

# Reducing visual cluttering

## Filtering data

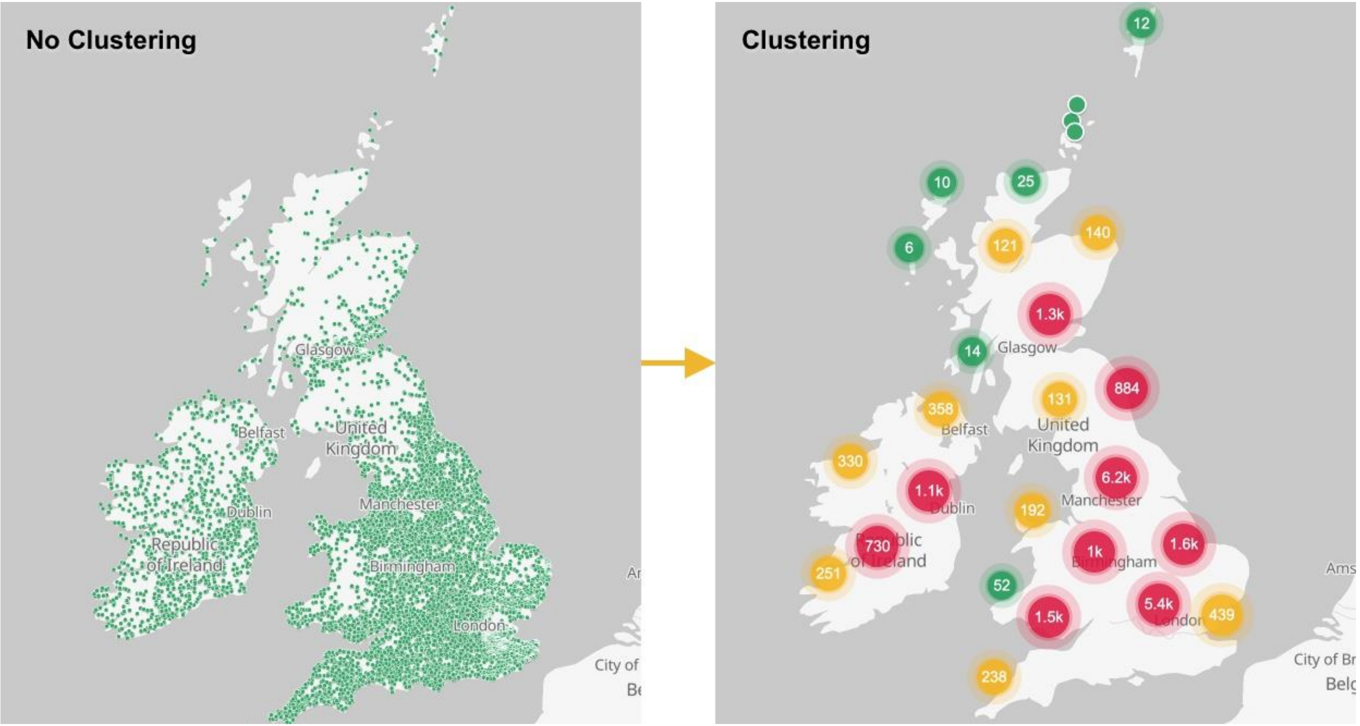



[Multivariate Network Exploration and Presentation: From Detail to Overview via Selections and Aggregations. van den Elzen, van Wijk, IEEE TVCG 20(12): 2014 (Proc. InfoVis 2014).]

# Reducing visual cluttering

## Aggregating data : Clustering

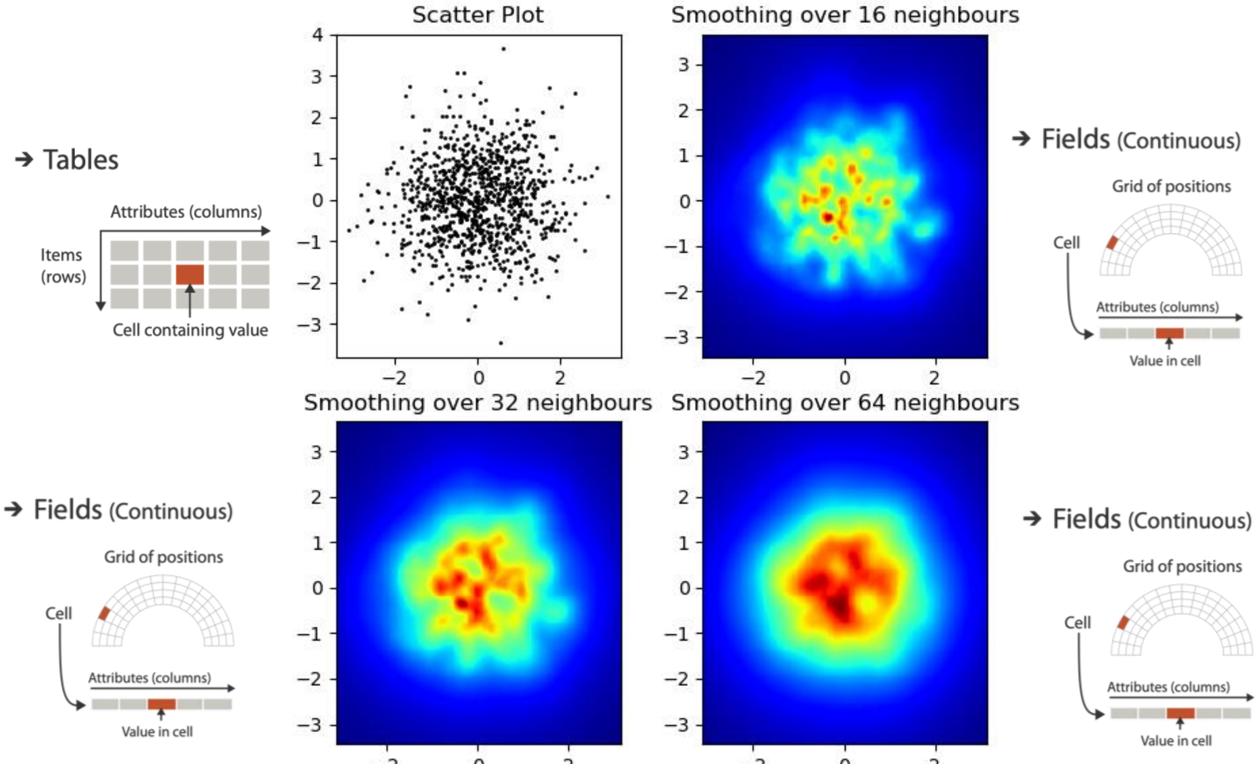



woosmap.com

# Reducing visual cluttering

## Aggregating data: Interpolating

Heatmap

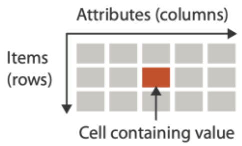



stackoverflow

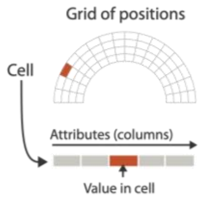
# Reducing visual cluttering

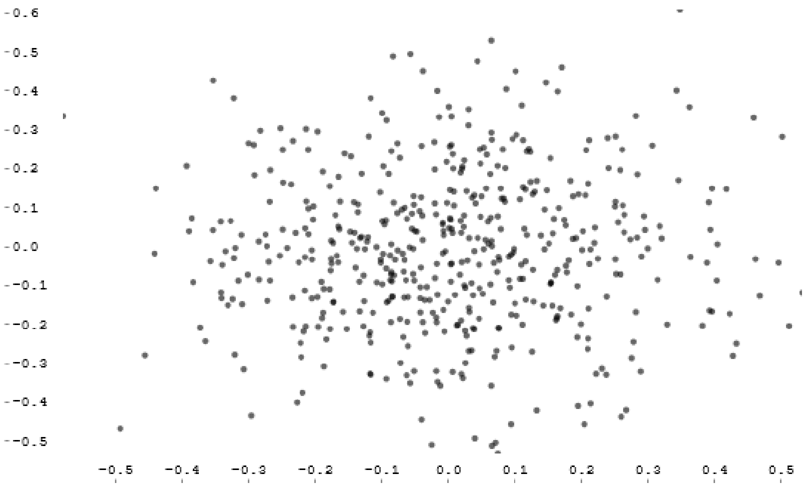
## Aggregating data: Binning

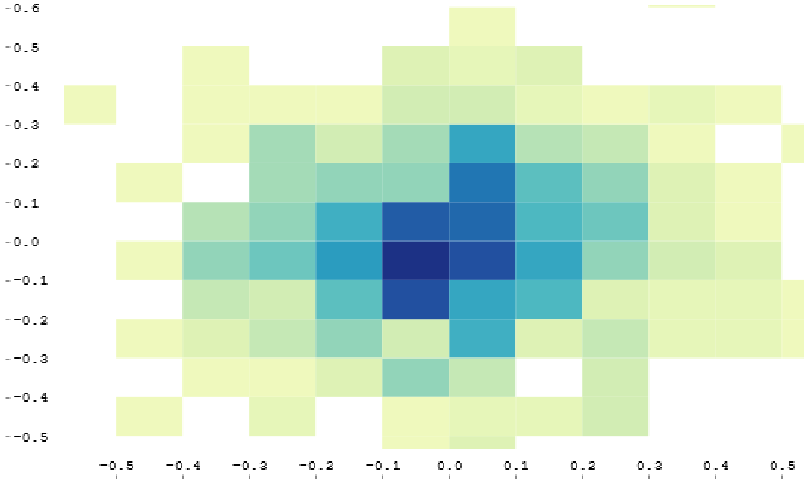
→ Tables




→ Fields (Continuous)



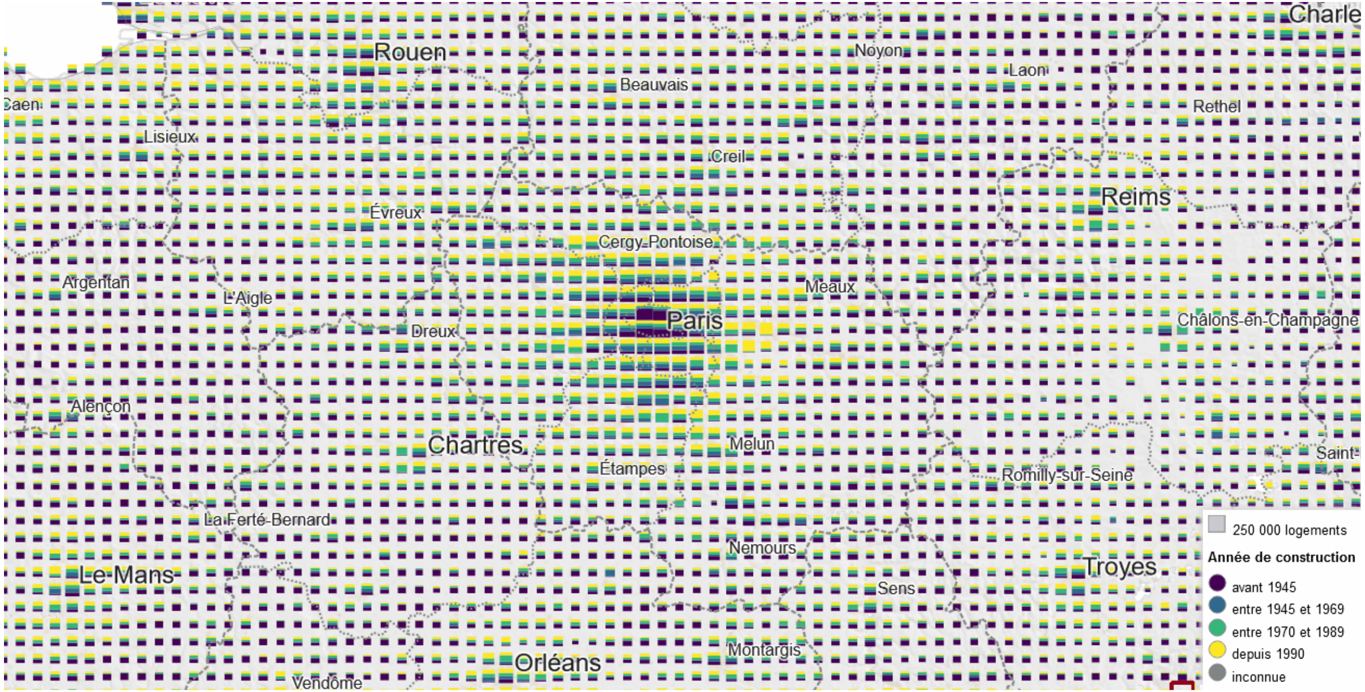



# Reducing visual cluttering

## Aggregating data: Binning

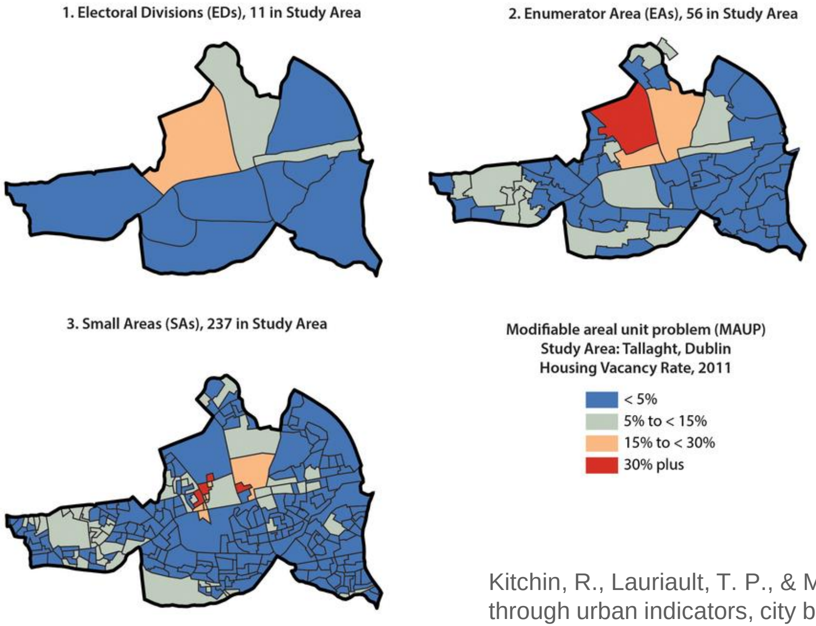
Binning according to scale




Gridviz

# Reducing visual cluttering

## Aggregating data: irregular grid




Be careful of biases which can be brought by irregular grids

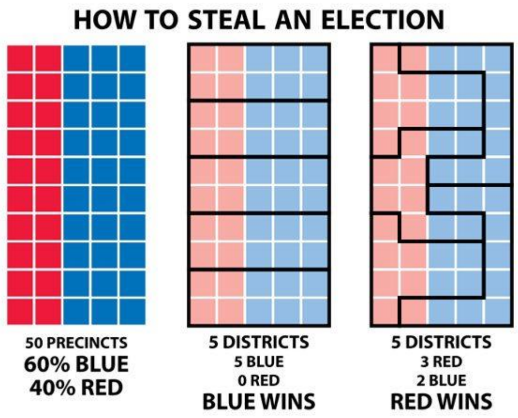
**MAUP (Modifiable Areal Unit Problem)**

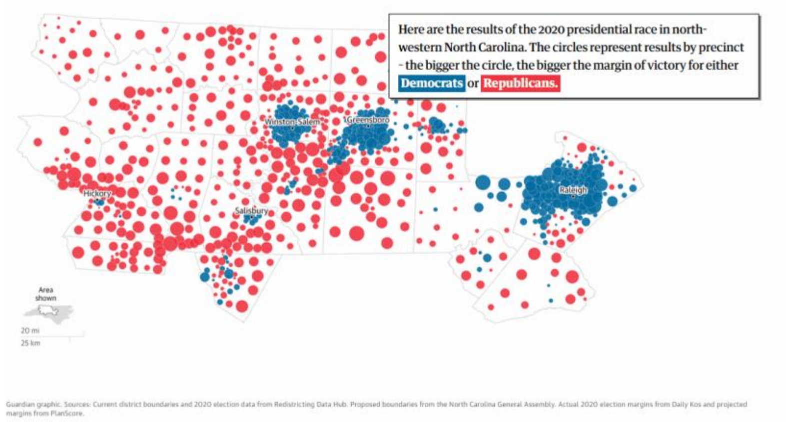
Kitchin, R., Lauriault, T. P., & McArdle, G. (2015). Knowing and governing cities through urban indicators, city benchmarking and real-time dashboards. *Regional Studies, Regional Science*, *2*(1), 6-28.

# Reducing visual cluttering

## Aggregating data: irregular grid

Example of MAUP effects: **the *gerrymandering***

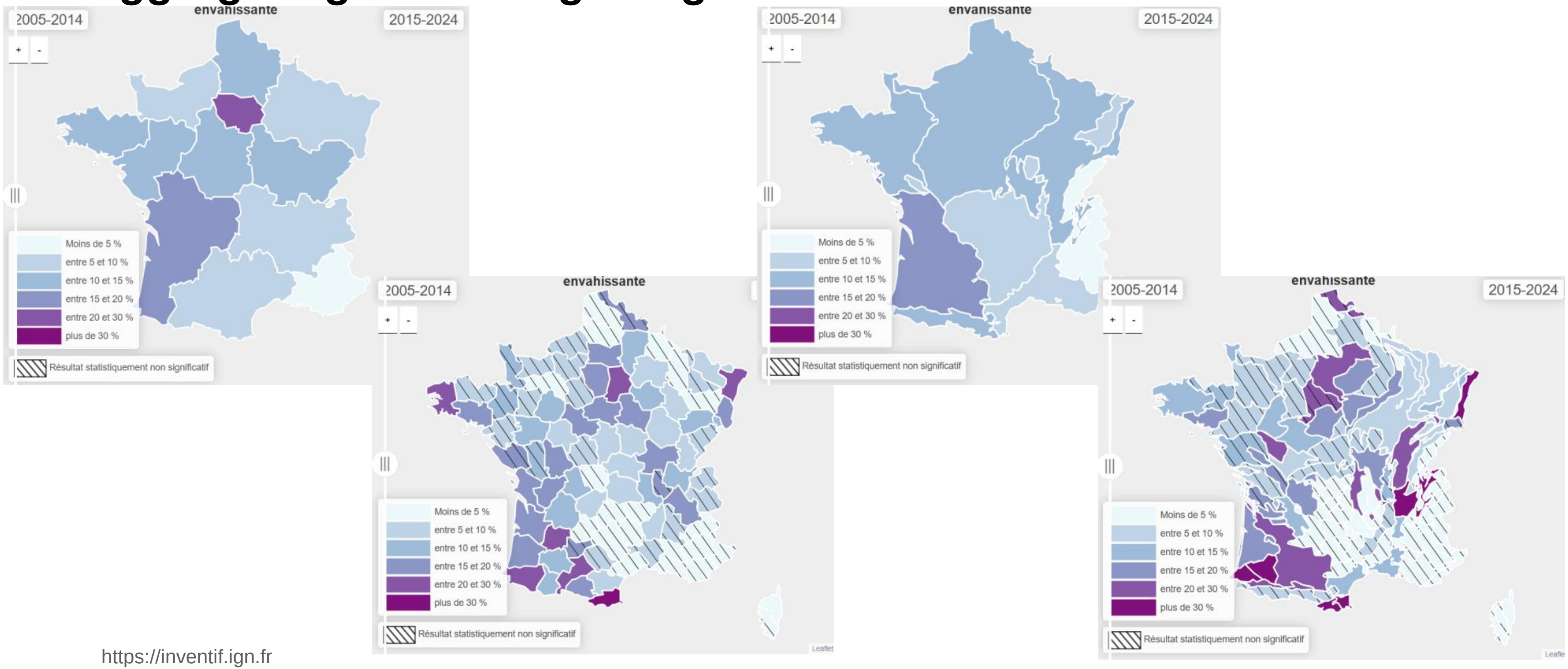




Wikipedia
WHQR
The Guardian

# Reducing visual cluttering

## Aggregating data: irregular grid

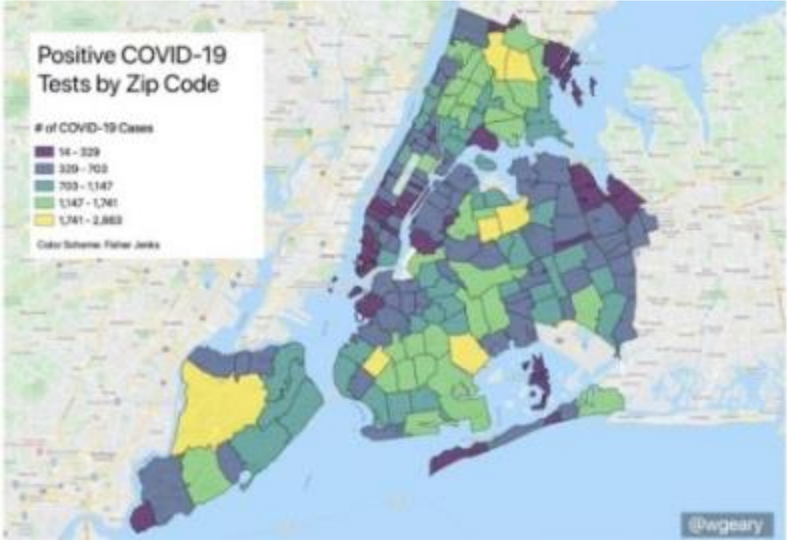
MAUP effects on forest data

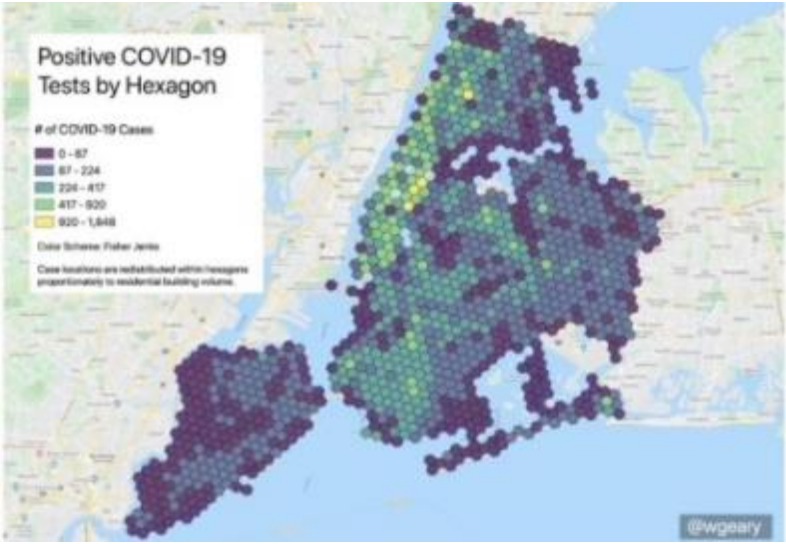



https://inventif.ign.fr

# Reducing visual cluttering

## Aggregating data: irregular grid



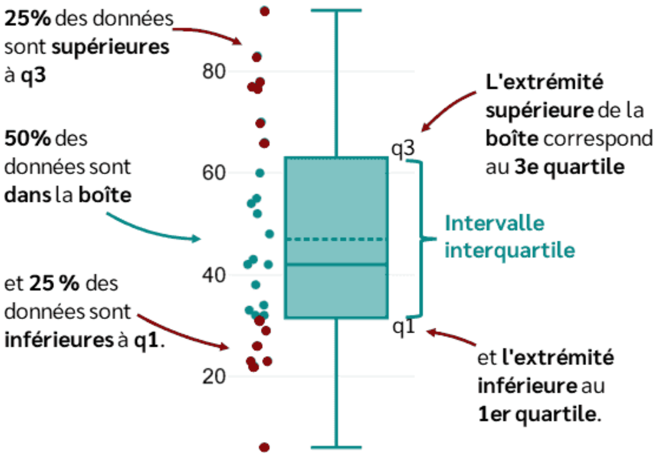




Will Geary, 2020

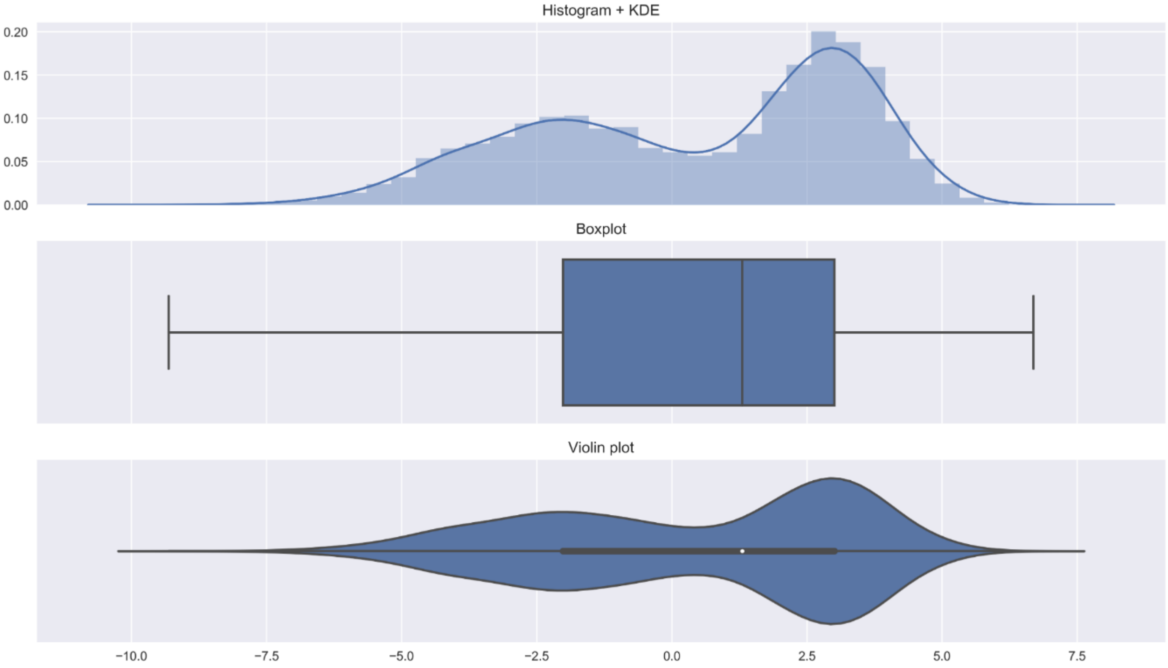
Regular grids can mitigate MAUP effects
Difficulty to identify geographic contours
Add difficulty for spatial contextualization

# Reducing visual cluttering

## Aggregating data : box plots and violin plots



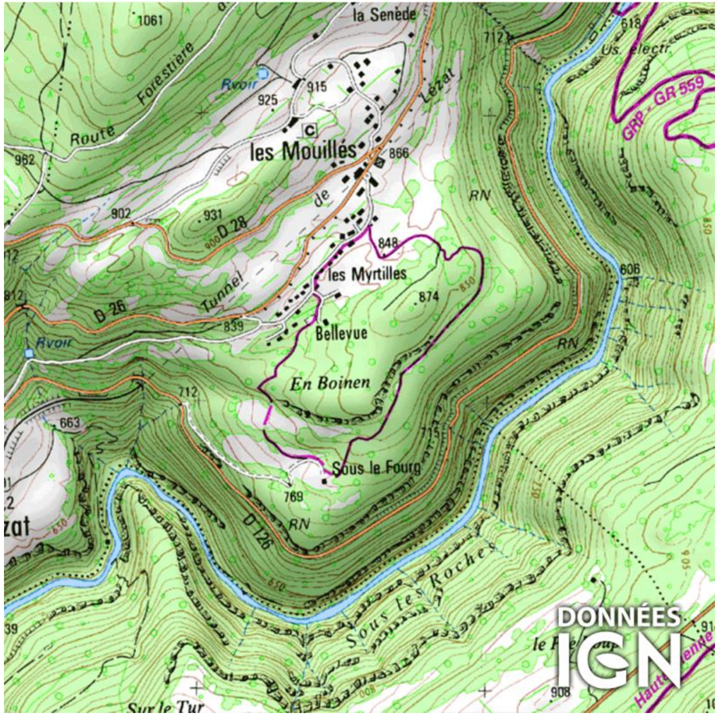

D3
towardsdatascience.com

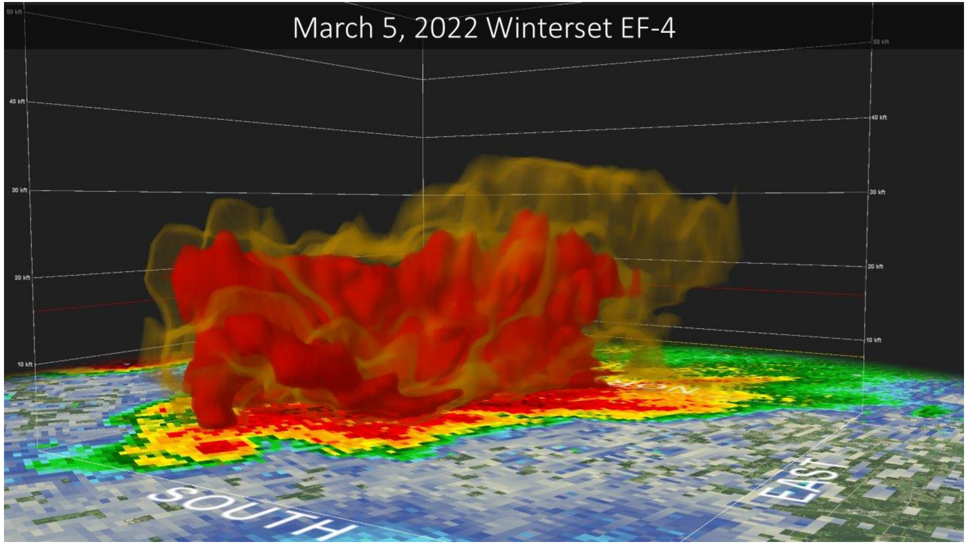


# Reducing visual cluttering

## Discrétisation: isocontours and isovolumes

DEM => altitude isolines



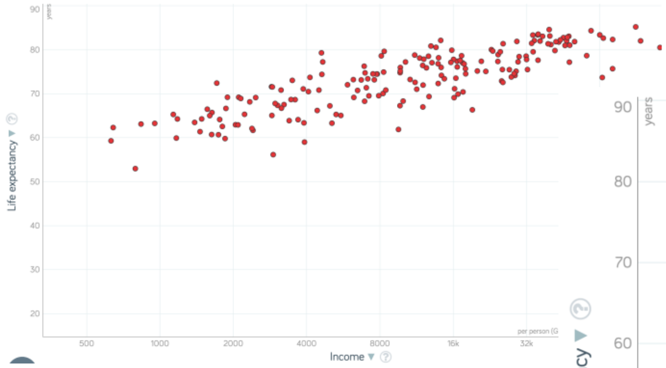




3D wind speed grid => isosurface

National Weather Service

# Multivariate representation

## Playing on multiple channels

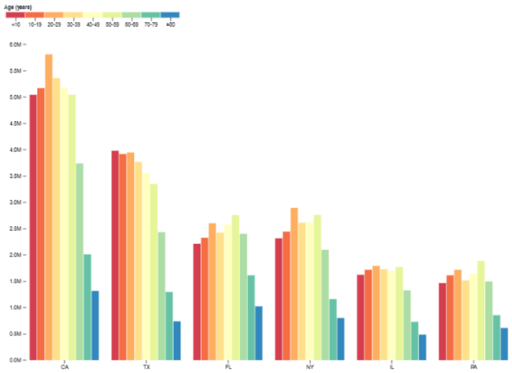




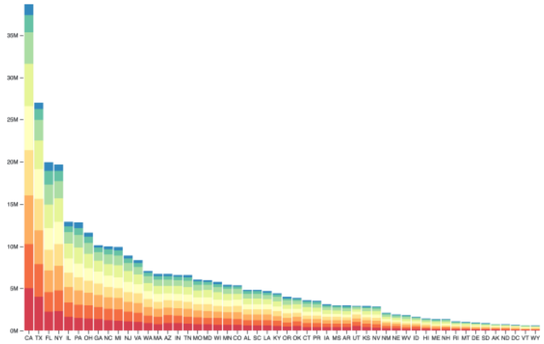
https://www.gapminder.org/

# Multivariate representation

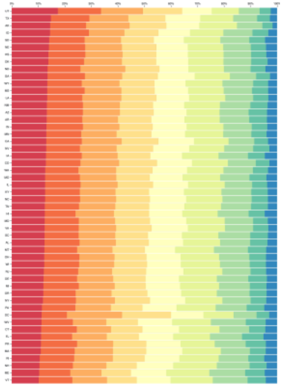
## Composed visualization: Embedding




Grouped bar charts

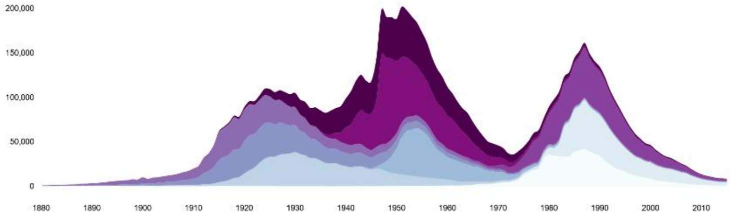


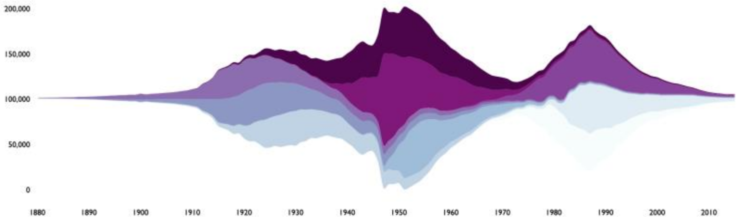
Stacked bar chart



Normalized stacked bar chart

Stacked Area chart - Steamchart

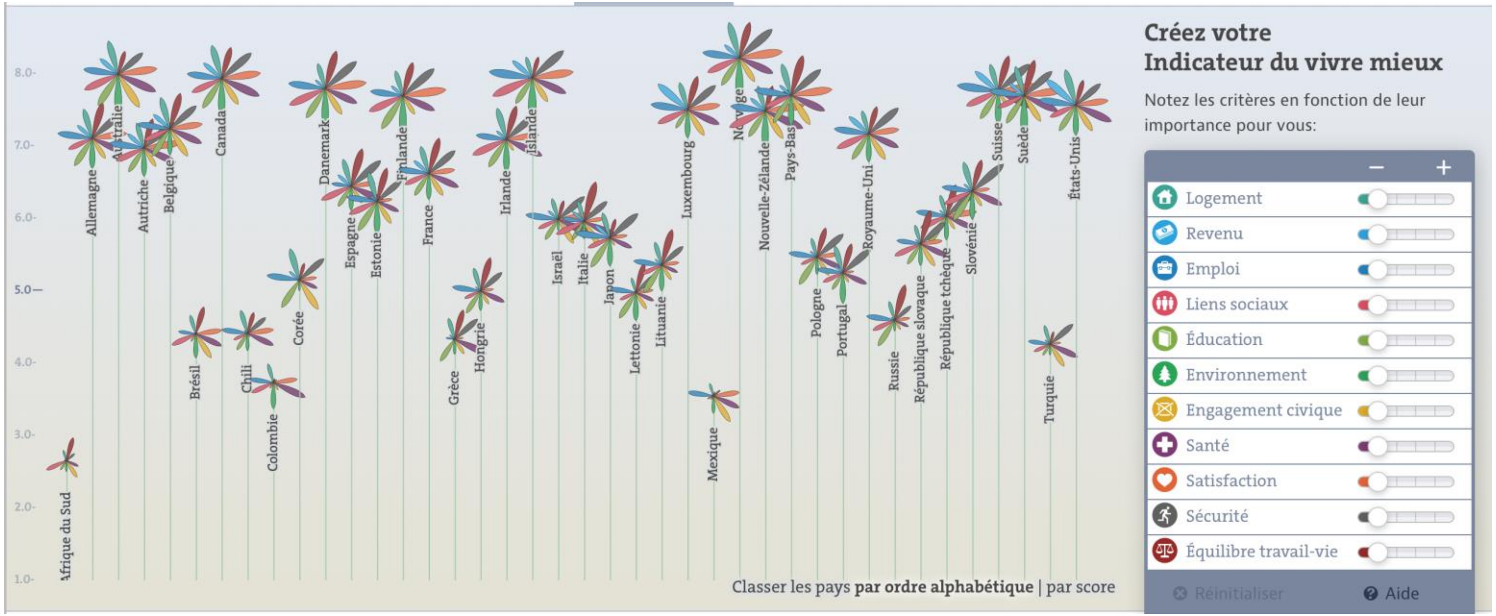


# Multivariate representation

## Composed visualization : embedding

**Glyphs (using symbols composed of several marks)**




http://www.oecdbetterlifeindex.org/fr/#/11111111111

# Multivariate representation

## Composed visualization : embedding

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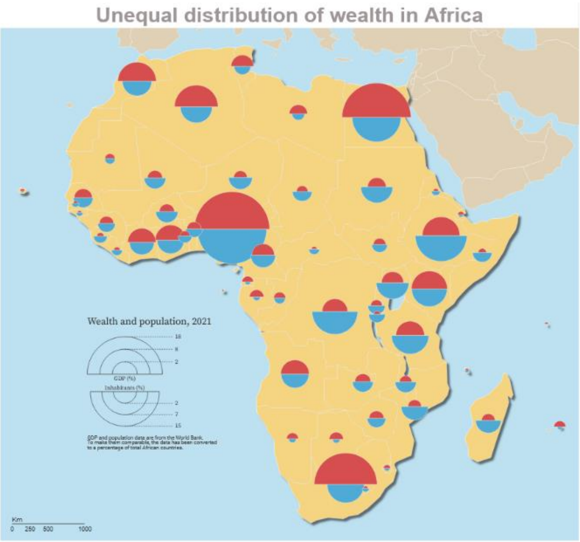
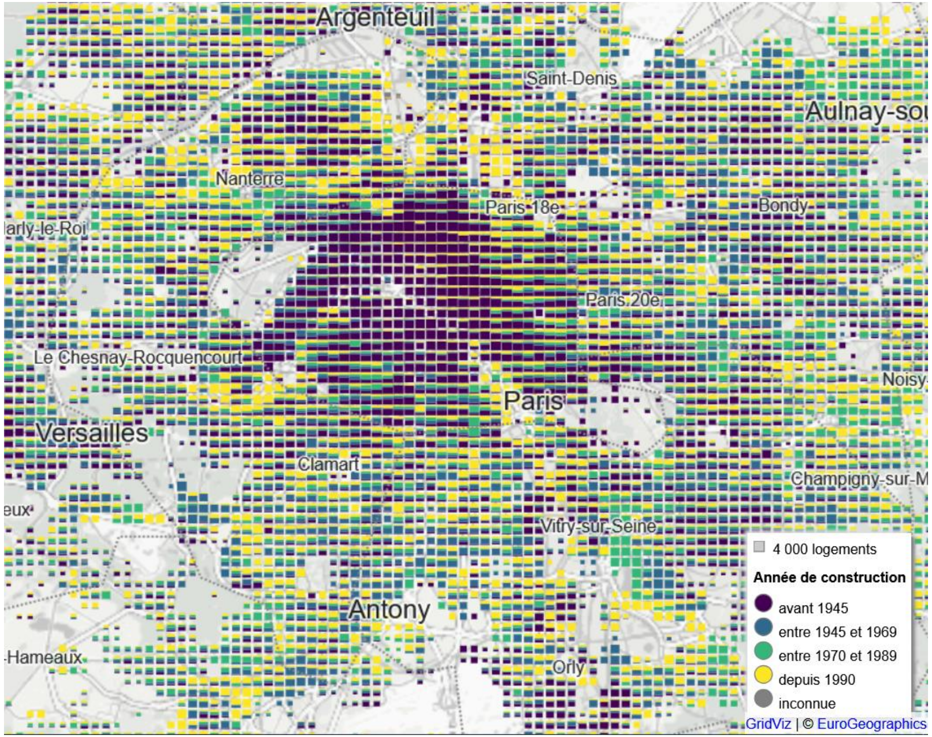



https://observablehq.com/@neocartocnrs/bertin-js-mashroom?collection=@neocartocnrs/bertin




Gridviz

# Multivariate representation

## Composed visualization – juxtaposition

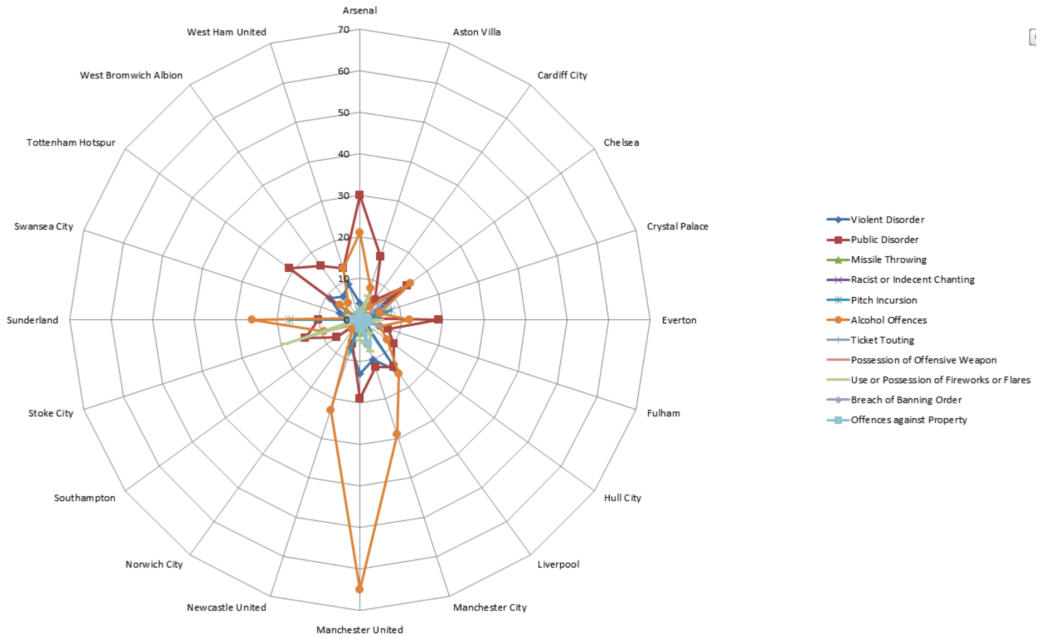

Small multiples (collection of representations)

Scatterplot Matrix

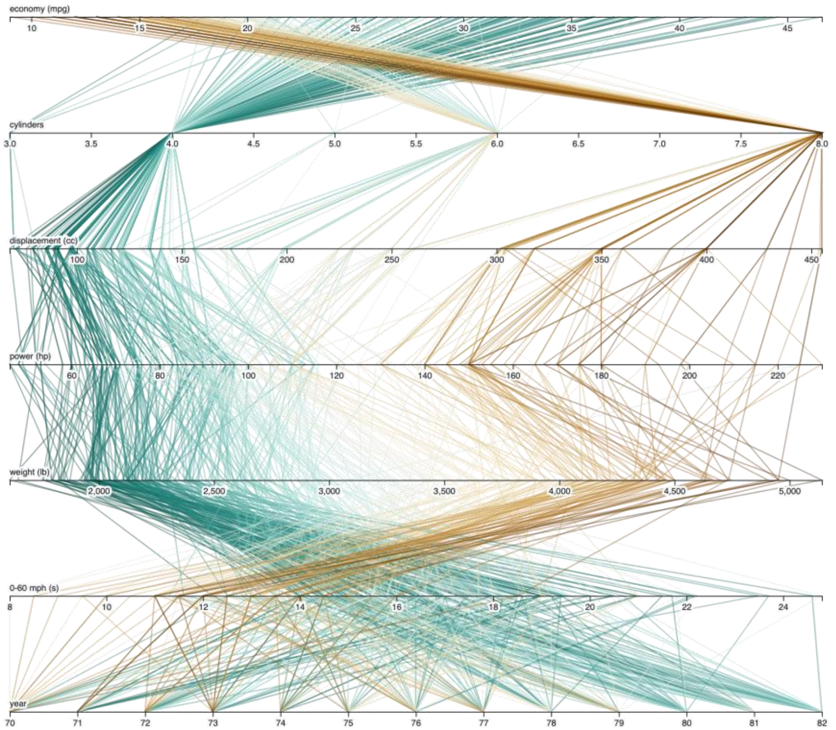
# Multivariate representation

## Composed visualization – juxtaposition

Parallel coordinates chart
https://www.gapminder.org/




Radar Chart
Munzner, 2014

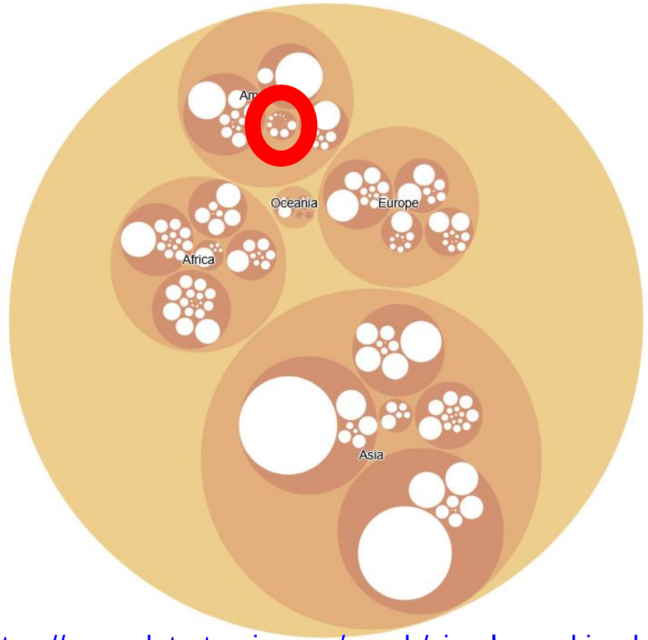


# Focus + context

Co-visualize..

..one detailed section of the dataset

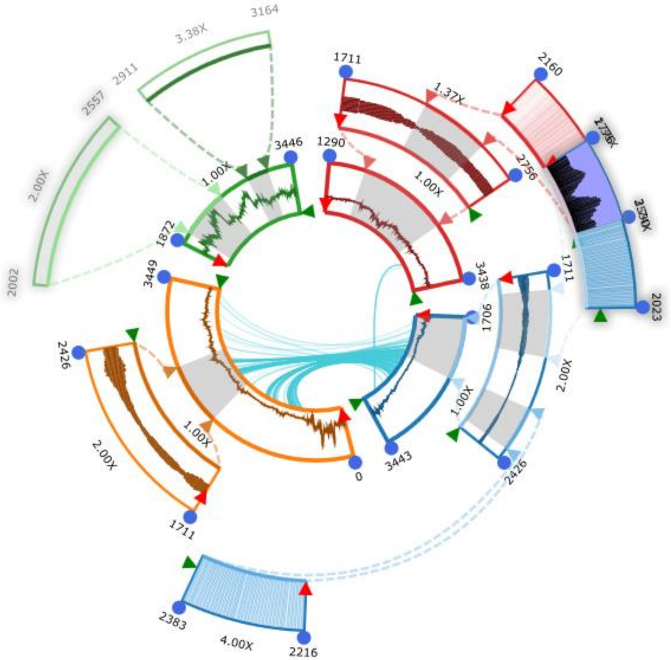
..dataset general distribution



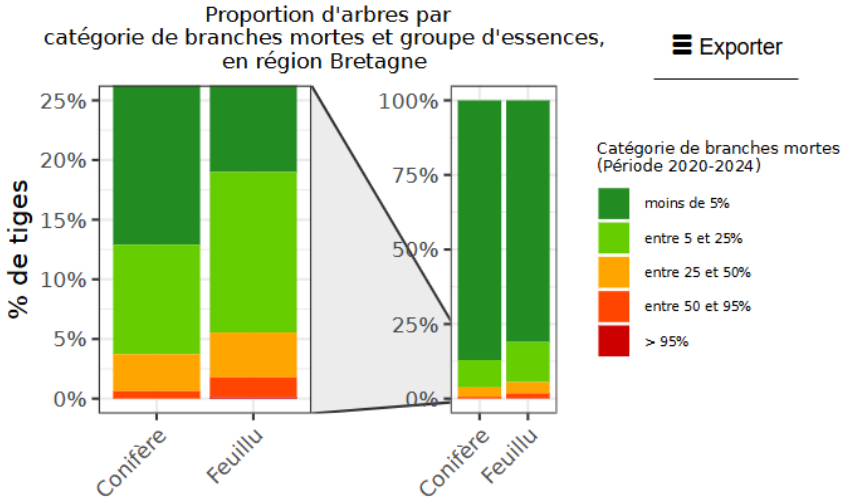

https://www.data-to-viz.com/graph/circularpacking.html

# Focus + context

## Embedding



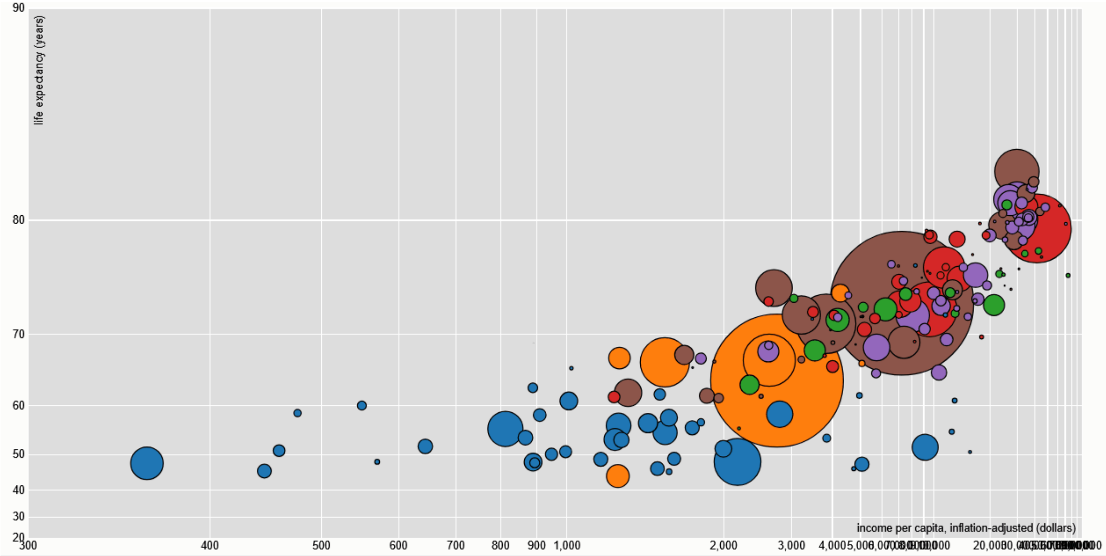
https://inventif.ign.fr/




Zhao, J., Chevalier, F., & Balakrishnan, R. (2011). KronoMiner: Using Multi-Foci Navigation for the Visual Exploration of Time-Series Data. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI), 1737–1746. https://doi.org/10.1145/1978942.1979195

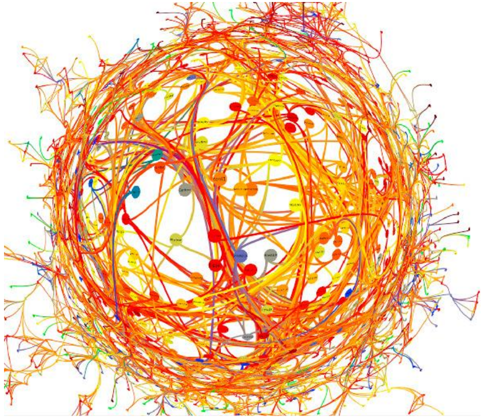
# Focus + context

## Spatial distorsion - lens

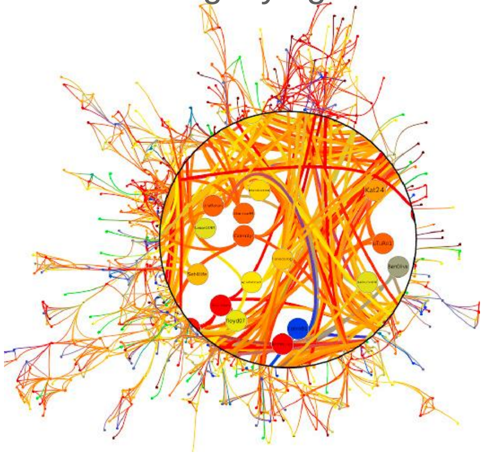



https://bost.ocks.org/mike/fisheye/

fisheye lens



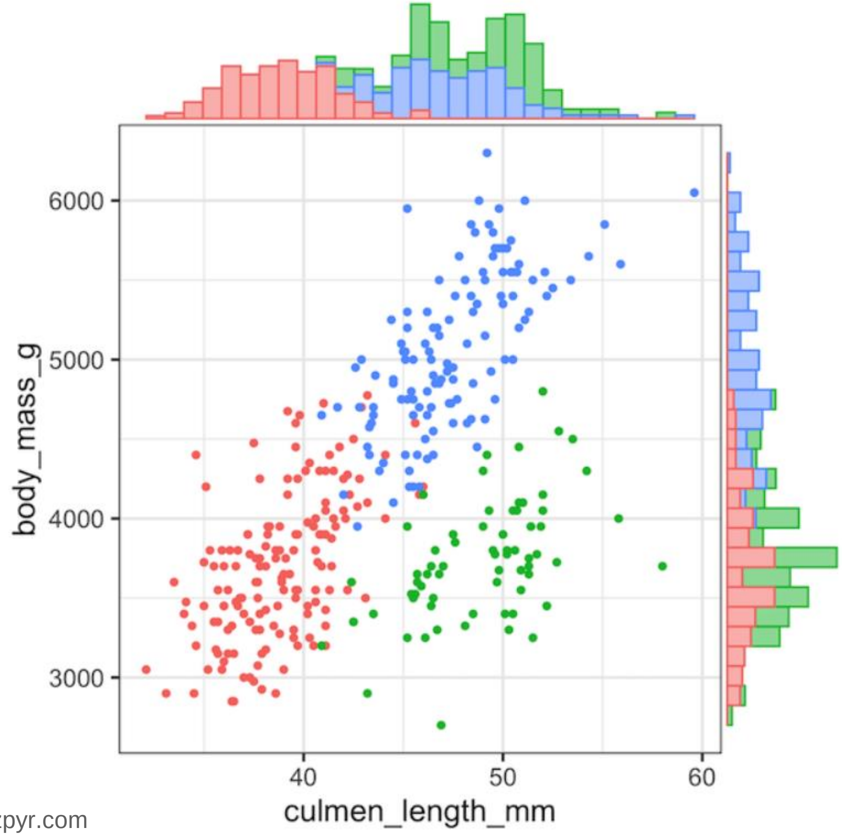
magnifying lens



*Living Flows: Enhanced Exploration of Edge-Bundled Graphs Based on GPU-Intensive Edge Rendering. Lambert, Auber, and Melançon. Proc. Intl. Conf. Information Visualisation (IV), pp. 523–530, 2010.*

# Focus + context

## Composed visualization
## Scatterplot + stacked bar chart



# Interaction

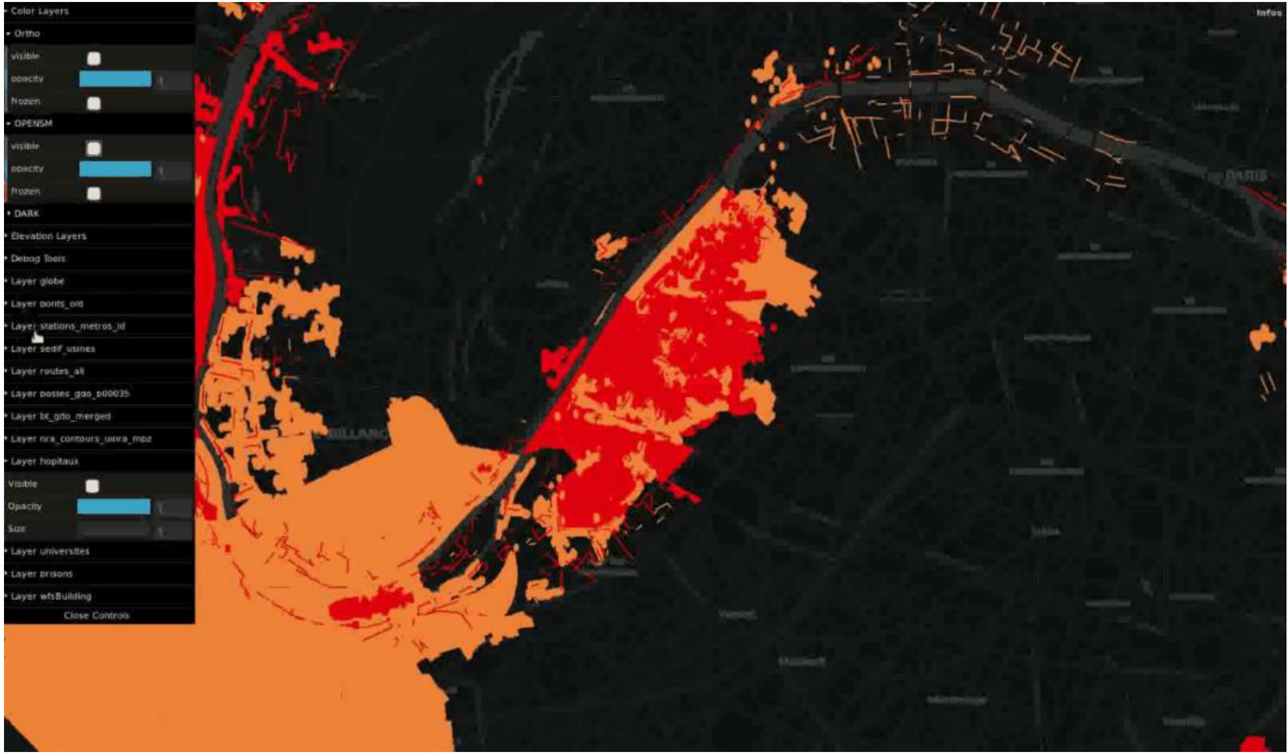
Navigate between

- ≠ scales
- ≠ styles
- Perspective (2D/3D)

Add additional data

Select data according to different dimension

Control animation




(iTowns project, A; Devaux, I. Lokhat 2016-2018)

# Interaction

## …for visualizing massive data

## Information Seeking Mantra

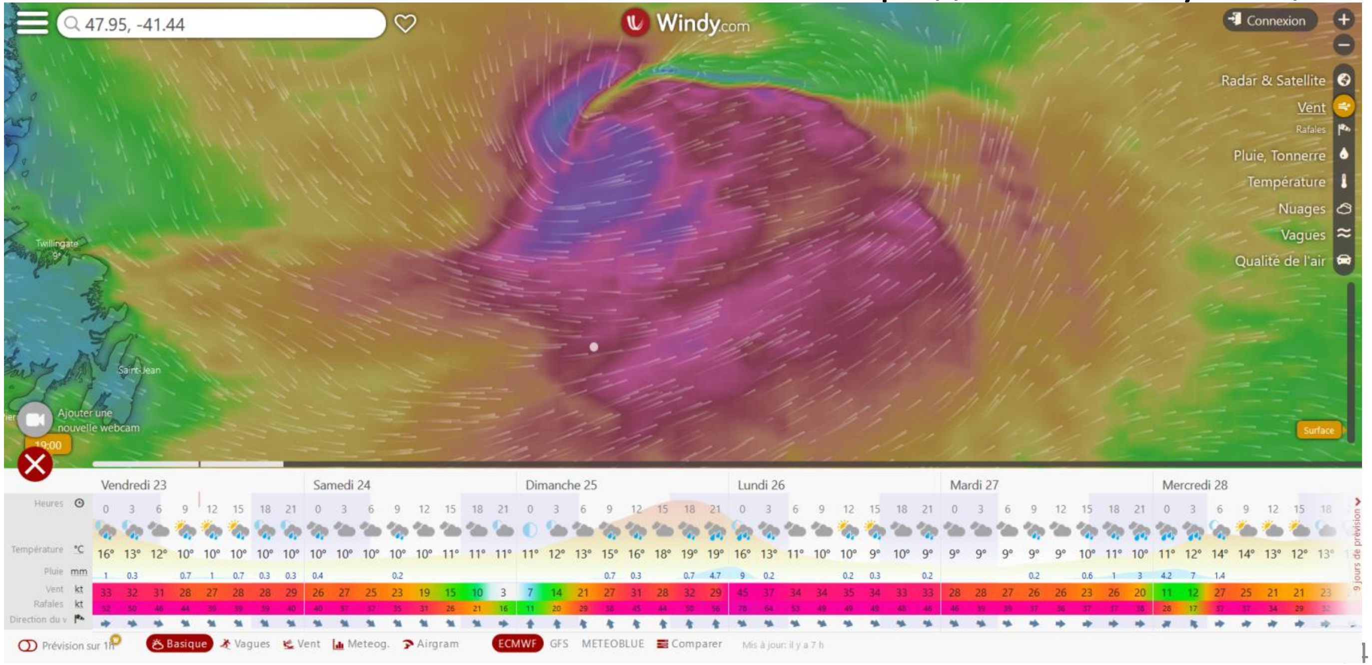
Shneiderman, 2003

https://www.windy.com/

*Overview first*

*Zoom and filter*

*Details on demand*



# 3D

## 3D geospatial data

Géoportail 3D

**2.5D**

X,Y,Z → 3D geographical space

**Only one Z coordinate** (ground altitude)

Enhancing topography representation

# 3D

## 3D geospatial data

Géoportail 3D

### 2.5D

X,Y,Z → 3D geographical space
**Only one Z coordinate** (ground altitude)

Enhancing topography representation


### 3D

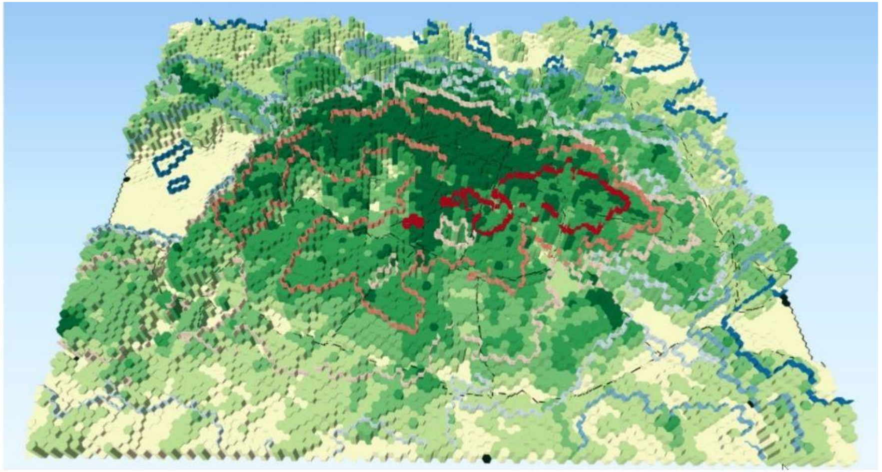
X,Y,Z → 3D geographical space
**Multiple Z coordinate** (altitude of ground, building, first floor, second floor, …)

3D urban model
3D data (t°, wind, etc.)

Google Earth VR

# 3D

2D geospatial data



**2D+1**

X,Y → geographical space
Z → non-spatial data

New graphic axis to represent data

Christophe, S., Gautier, J., Chapron, P., Riley, L., & Masson, V. (2022). 3D geovisualization for visual analysis of urban climate. *Cybergeo: European Journal of Geography*.

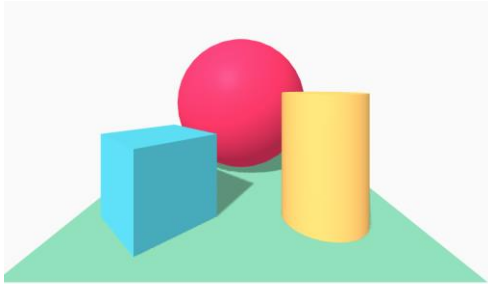

**Space-time cube (2D+1)**

X,Y → geographical space
Z → time

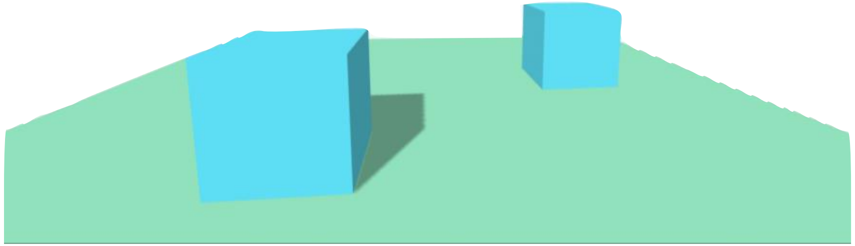
Gautier, Lobo et al., 2021

# Problems related to 3D perception

- Occlusion:



- Relative size:



# Problems related to 3D perception

- Relative density:

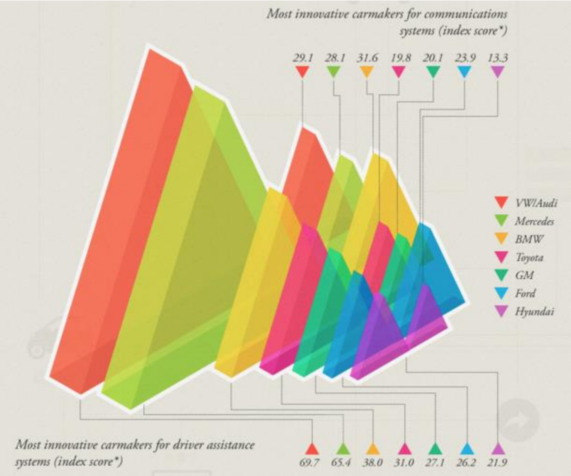
Christophe Libert ; FreeImages

# Problems related to 3D perception

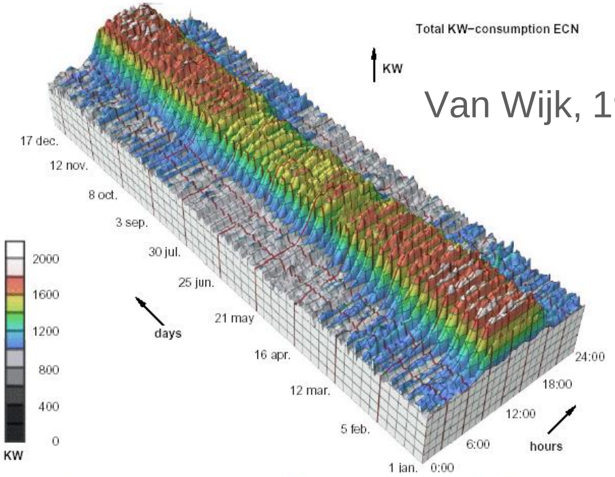
- Relative height:

Max Mitenkov ; FreeImages

# Avoid unjustified 3D

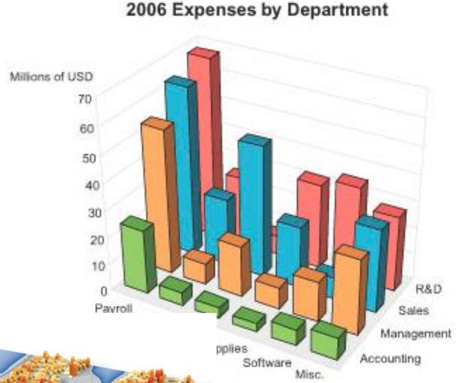


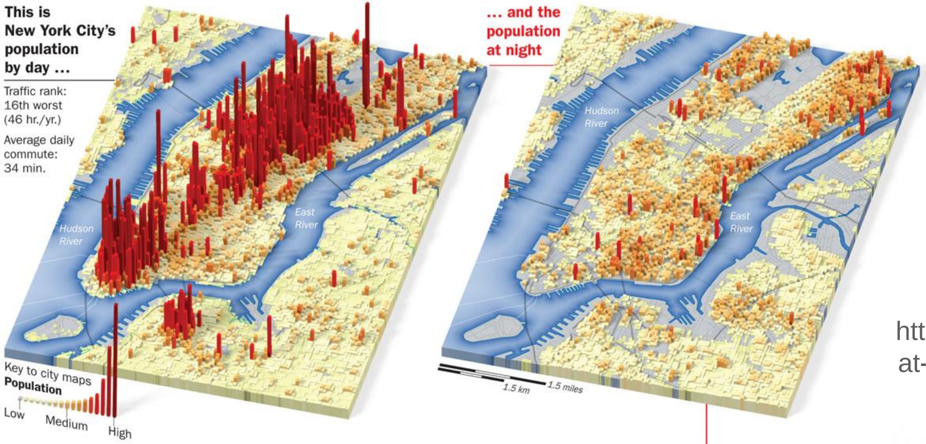

http://viz.wtf/post/137826497077/eye-popping-3d-triangles




Van Wijk, 1999

[http://perceptualedge.com/files/GraphDesignIQ.html]




https://www.6sqft.com/what-nycs-population-looks-like-day-vs-night/

# Avoid unjustified 3D

**2D is often better than 3D**

3D generaly implies longer analysis time and more important cognitive efforts for the user [Munzner 2014]

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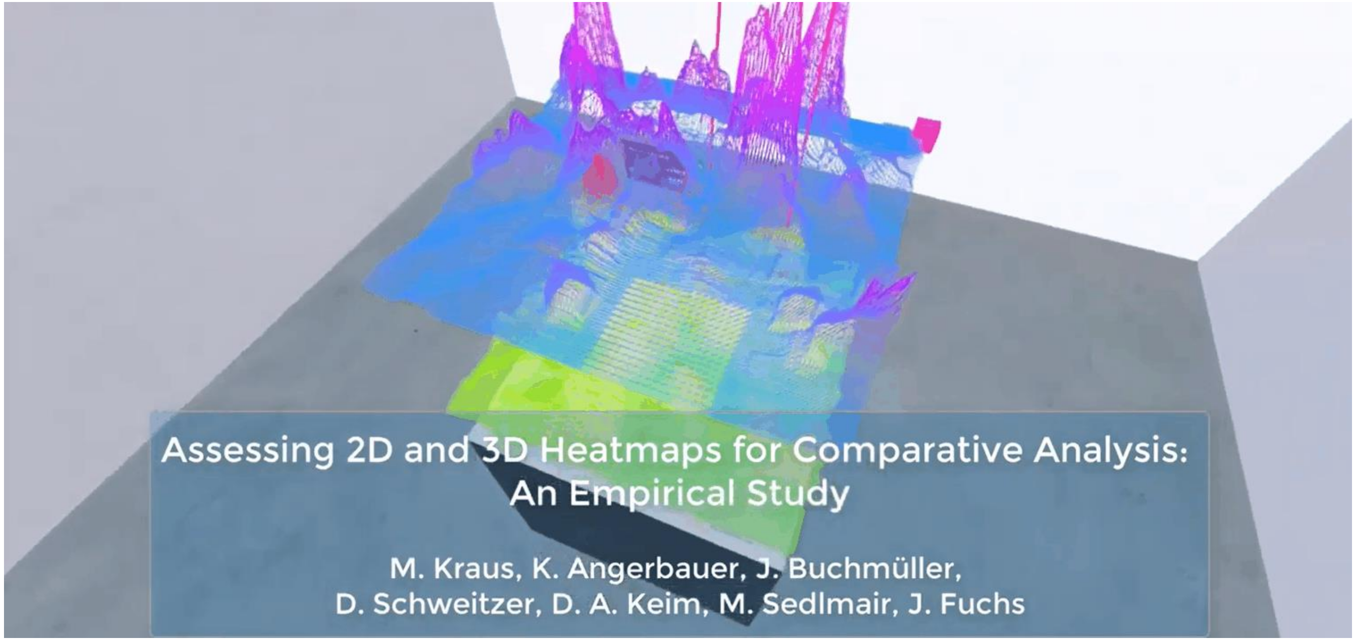
**3D is justified when there is a benefit to make perceive shapes and distribution in a 3D space**

→ 3D spatial or geospatial data

3D representation of statistical data (2D+1 or 3D chart) is **controversial**

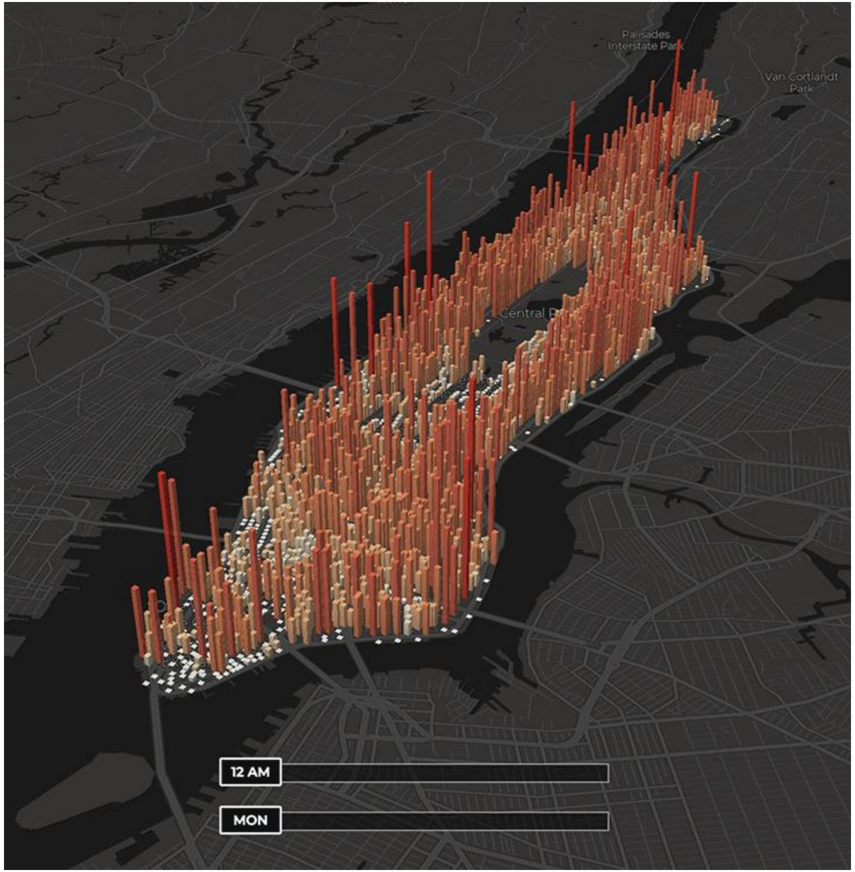
# Unjustified 3D?




Kraus, M., Angerbauer, K., Buchmüller, J., Schweitzer, D., Keim, D. A., Sedlmair, M., & Fuchs, J. (2020, April). Assessing 2D and 3D Heatmaps for Comparative Analysis: An Empirical Study. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (pp. 1-14).

# Unjustified 3D?

Impactful visualization for a large public




https://www.weforum.org/stories/2018/06/animation-the-population-pulse-of-a-manhattan-workday/

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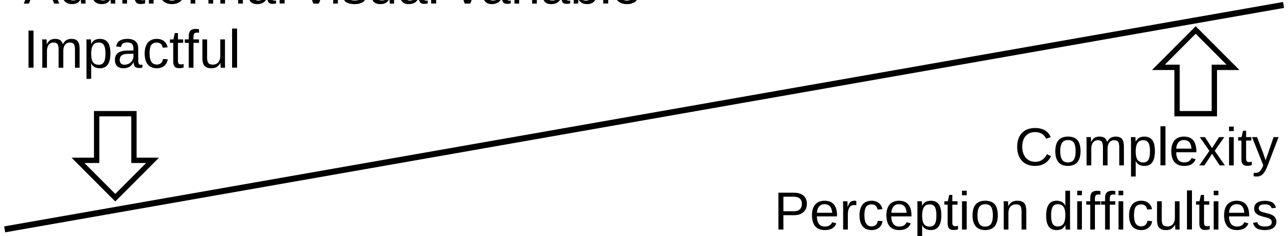


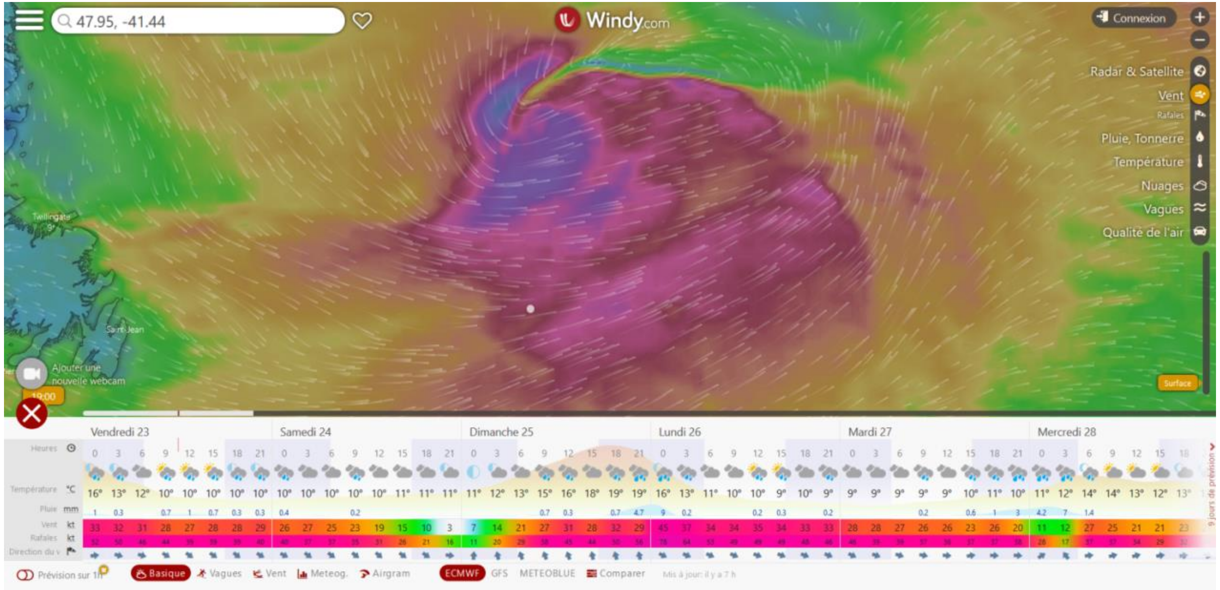
# Avoid unjustified 2D as well

## For some tasks, a simple table is better than a 2D visualization

| Country | Total confirmed deaths due to COVID-19 per million people deaths per million | GDP growth from previous year, 2020 Q2 percent |
|---|---|---|
| Austria | 81.39 | -13.3% |
| Belgium | 853.44 | -14.5% |
| Bulgaria | 87.07 | -8.2% |
| Canada | 241.45 | -13.5% |
| Chile | 584.90 | -13.7% |
| China | 3.28 | 3.2% |
| Colombia | 374.66 | -15.7% |
| Cyprus | 23.98 | -11.9% |
| Czech Republic | 39.31 | -10.7% |
| Denmark | 107.73 | -8.5% |
| Finland | 60.46 | -5.2% |
| France | 468.83 | -19.0% |
| G7 | | -12.1% |
| Germany | 110.94 | -11.7% |
| Hungary | 63.56 | -13.5% |
| Indonesia | 26.55 | -5.4% |
| Israel | 104.67 | -7.8% |
| Italy | 586.70 | -17.3% |
| Japan | 9.99 | -10.0% |
| Latvia | 18.03 | -9.6% |

https://ourworldindata.org

*Ex: details on demand*




https://www.windy.com/

# Conclusion

- Objectif de votre visualisation ?

  *Explorer des données, communiquer une information…*
- Complexité du phénomène à représenter ?

  *Nature et dimensions de vos données, incertitude, structure du jeu de données…*
- Niveau d'expertise de l'utilisateur ?

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- Niveau d’expertise de l’utilisateur ?

⇒ 2D ou 3D? Animation? Interaction? À quel degré? Représentation cartographique? Quel style? Diagrammes? Le(s)quel(s)? Fil narratif ?

# Conclusion

- Objectif de votre visualisation ?

  *Explorer des données, communiquer une information…*
- Complexité du phénomène à représenter ?

  *Nature et dimensions de vos données, incertitude, structure du jeu de données…*
- Niveau d’expertise de l’utilisateur ?

⇒ 2D ou 3D? Animation? Interaction? À quel degré? Représentation cartographique? Quel style? Diagrammes? Le(s)quel(s)? Fil narratif ?

- Available time, future of the visualization, programming skills

# Conclusion

- Objective of your visualization?

    *Explore data, communicate an information…*
- Complexity of the represented phenomena?

    *Nature, structure, uncertainty, dimensions of your data…*
- Level of expertise of end-users?

⇒ 2D or 3D? Animation? Interaction? To which extent? Cartographic representation? Which style? Charts? Which one(s)? Narrative patterns?

- Available time, future of the visualization, programming skills

⇒ Which technical solutions?

# Conclusion

Different approaches, no global solution.

→ *Always start from your end-user use case, needs, and habits*

The main objective is still a functionnal tool, not necessarely a beautiful one :

***Function first, form next***

*Munzner 2014*

# Practical exercise with Vega-Lite